

Lecture 13 Image Reconstruction

(Diffusion Models for Medical Imaging)

Yuyao Zhang PhD

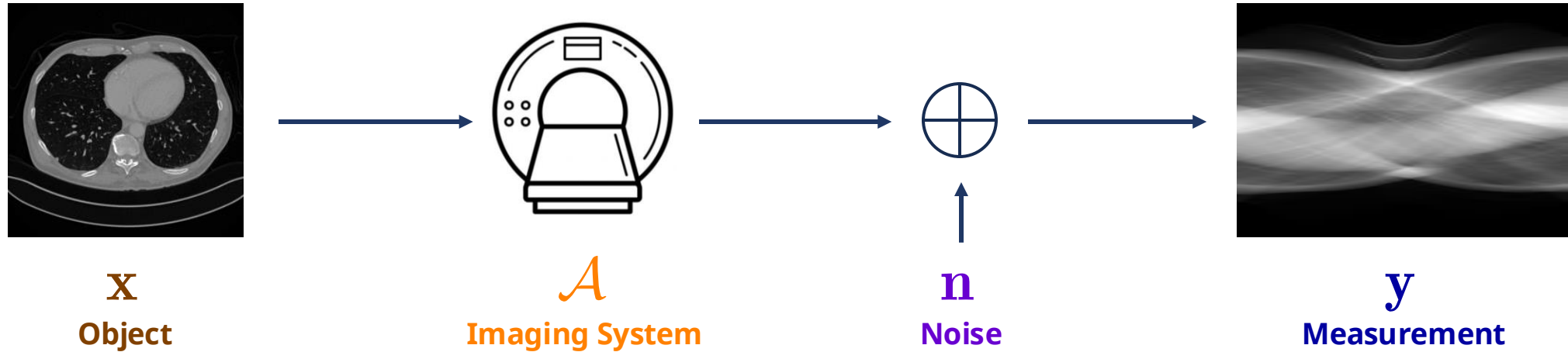
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SIST Building-3 420

Outline

1. Medical Imaging Inverse problem and diffusion models
2. Major Paradigms of IP solving with DMs
3. Challenge Problem in medical imaging
 - 3D problem
 - Fuzzy forward model
 - Out-of-distribution

Medical Imaging inverse problem



Medical imaging systems cast as the forward model

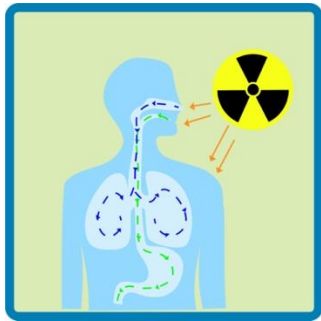
$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$$

- Acquiring the image : **inverse problem**
- Ex. **CT, MRI**, etc.

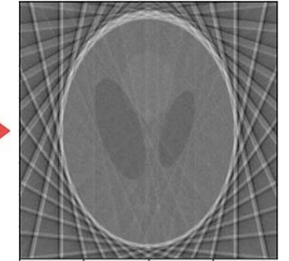
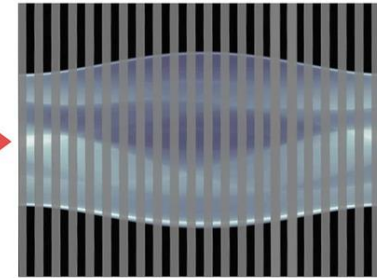
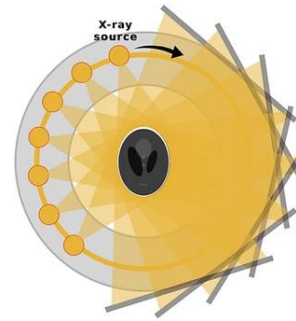
| Background: *CT/MRI Acquisition*

Challenging in CT ?

□ *Radiation Exposure*



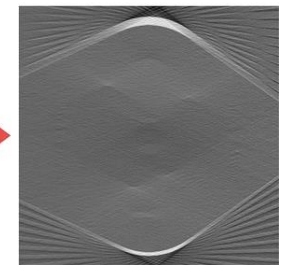
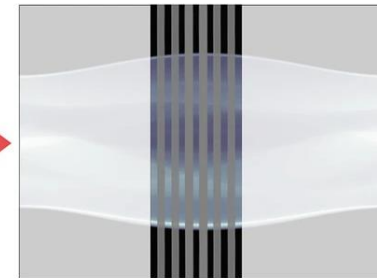
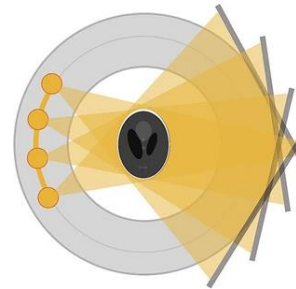
→ **Sparse-view
CT (SVCT)** →



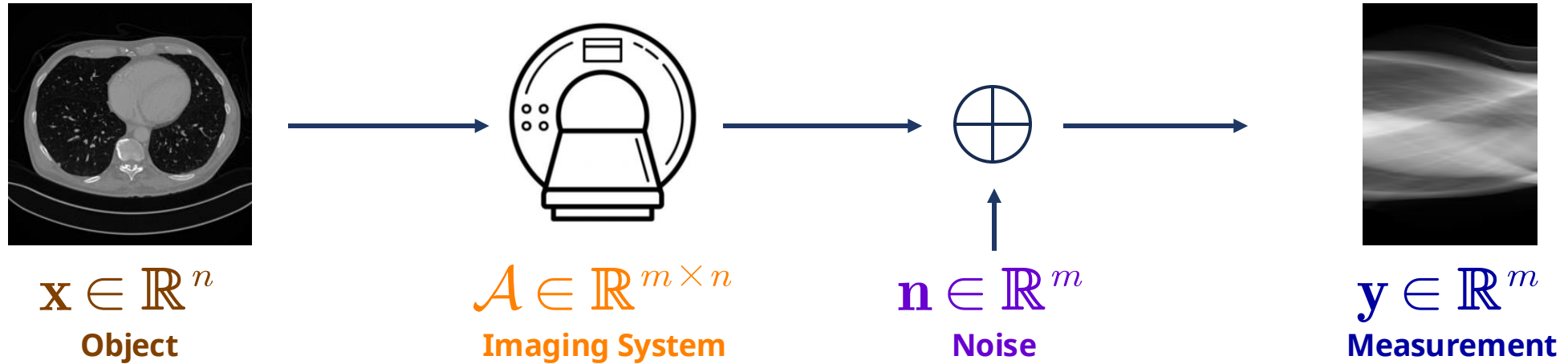
□ *Special Instrument Limitations (C-arm)*



→ **Limited-angle
CT (LACT)** →



Medical Imaging inverse problem



But in some real-world scenarios?

when $m \ll n$?

- Acquiring the image : **ill-posed inverse problem**
- Ex. **LACT, SVCT, CS MRI**, etc.

Medical Imaging inverse problem

Bayesian inference for inverse problems

$$\mathbf{y} = \mathcal{A}(\mathbf{x}) + \mathbf{n}$$

A Maximum-a-posteriori (MAP) estimation

$$\hat{\mathbf{x}} = \underset{\mathbf{x}}{\operatorname{argmax}}$$

Posterior

$$p(\mathbf{x} | \mathbf{y})$$

$$= \underset{\mathbf{x}}{\operatorname{argmin}}$$

$$-\log p(\mathbf{y} | \mathbf{x})$$

Likelihood

$$-\log p(\mathbf{x})$$

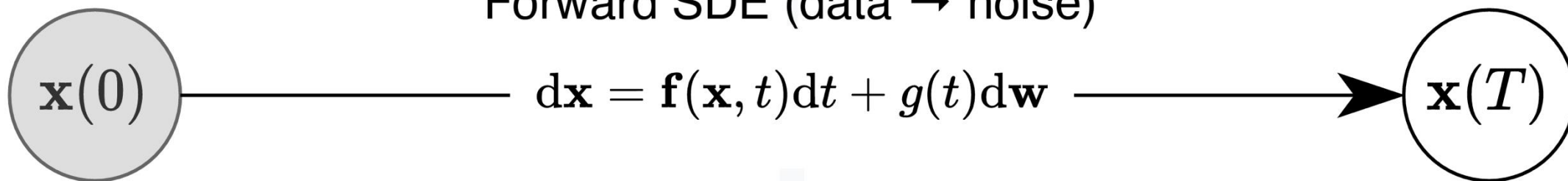
Prior

$$\|\mathbf{y} - \mathcal{A}(\mathbf{x})\|^2$$

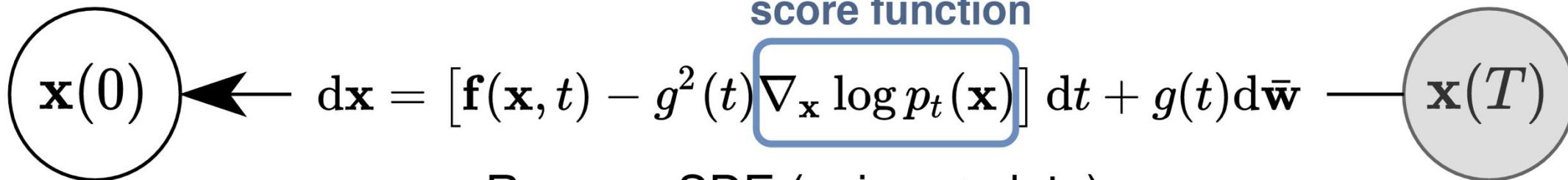
- TV
- Low-rank
- CNN
- GAN
- Diffusion Model
- ...

Learning the Prior of medical images via diffusion models

Forward SDE (data $\rightarrow$ noise)



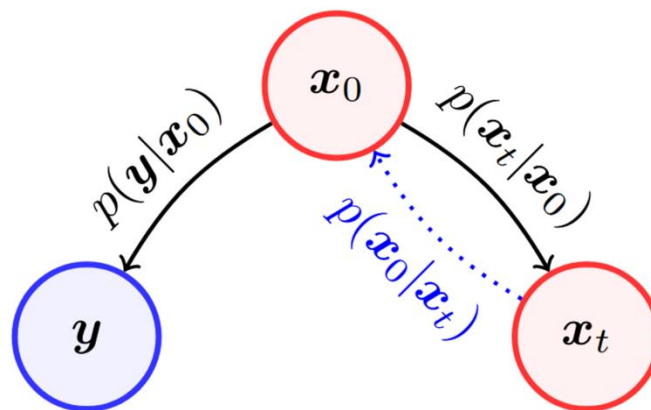
score function



Reverse SDE (noise $\rightarrow$ data)

Learning the Prior of medical images via diffusion models

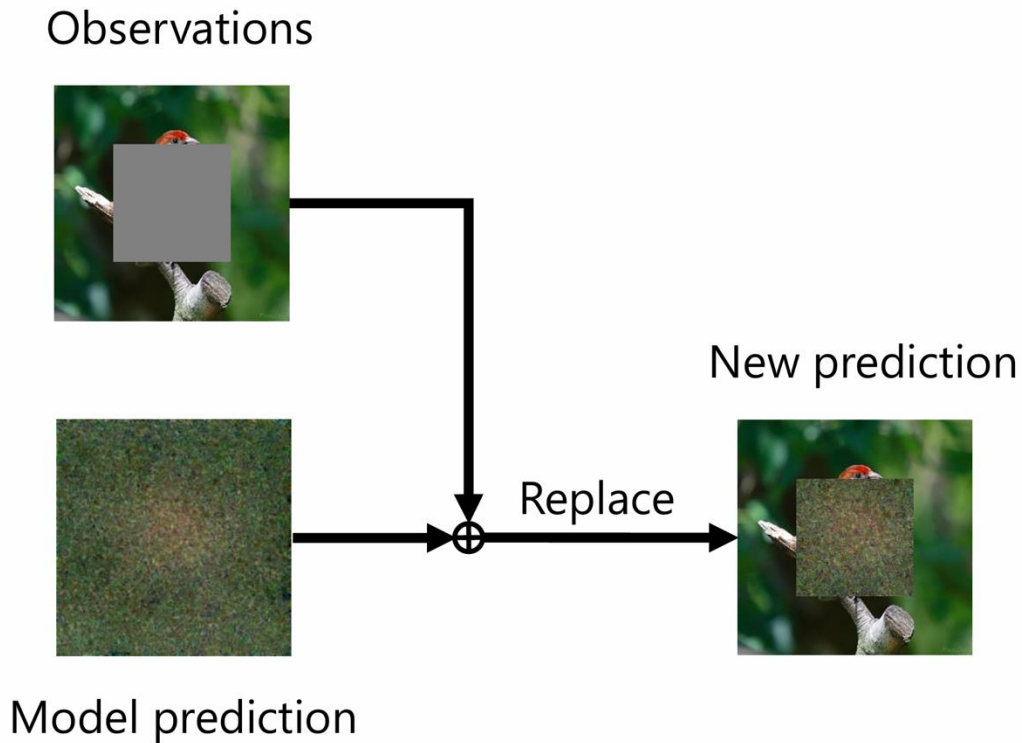
$$\underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t | \mathbf{y})}_{\text{"Score"}} = \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)}_{\text{Prior diffusion model}} + \underbrace{\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y} | \mathbf{x}_t)}_{\text{This is not known!}}$$



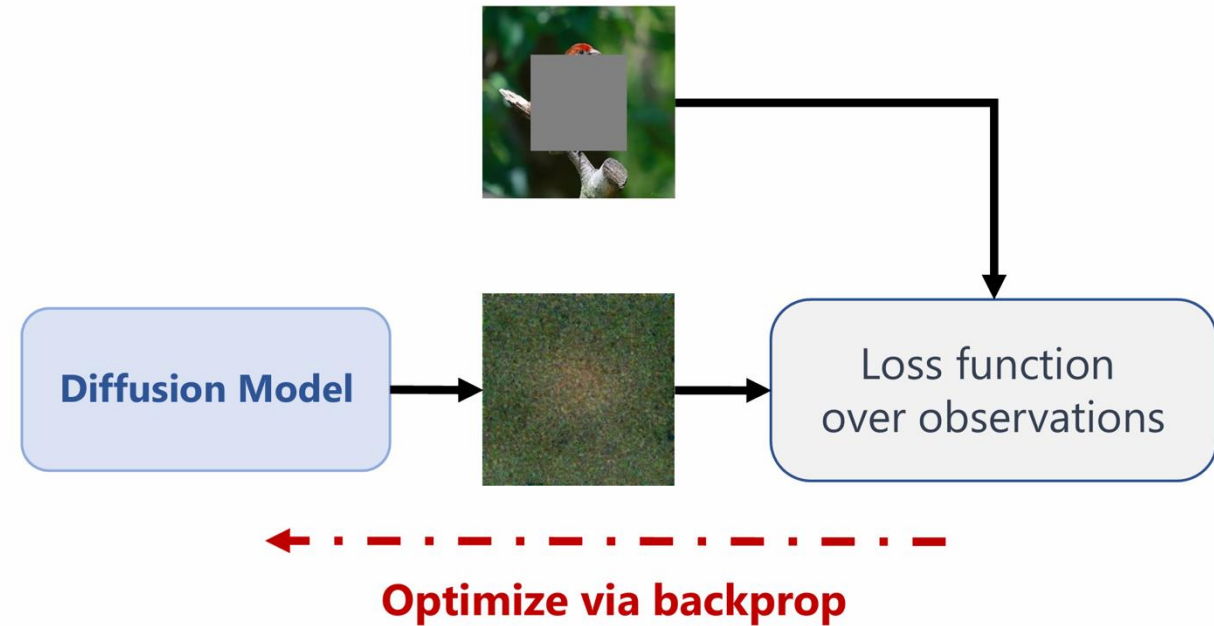
Diffusion models for Medical Imaging inverse problem

Two major Paradigms for computing likelihood term

Replacement-based



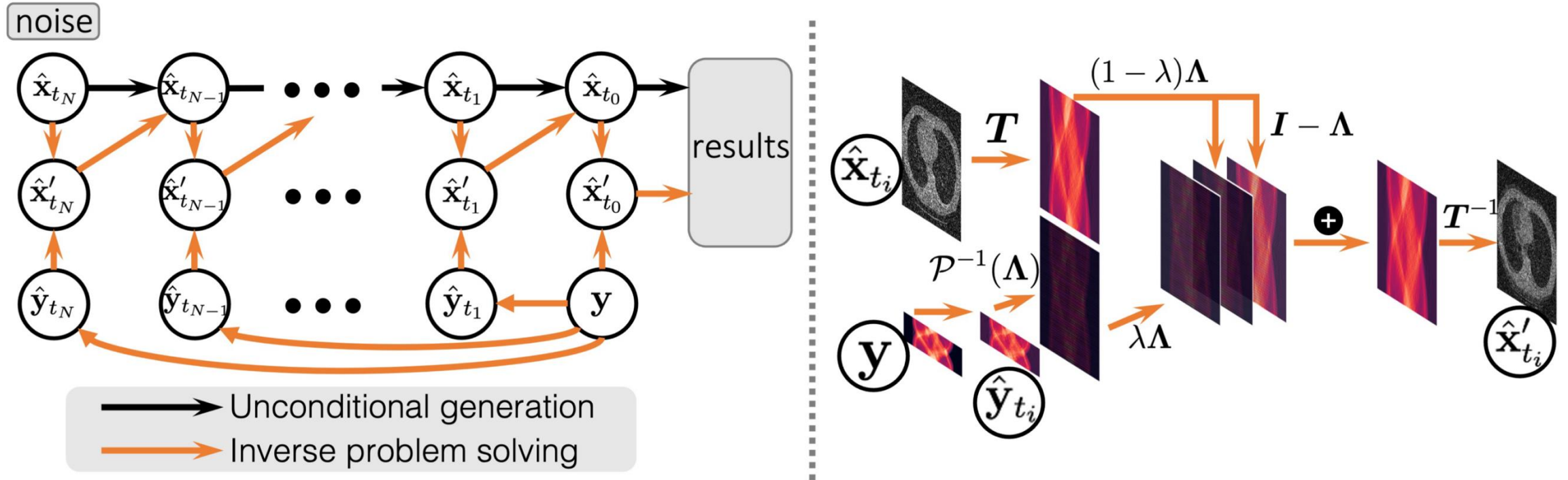
Reconstruction-based



Diffusion models for Medical Imaging inverse problem

Score-SDE, Song et al., 2022

Replacement-based



$$\alpha(t)\mathbf{y} + \beta(t)\mathcal{A}(\mathbf{z}) \rightarrow \mathbf{y}_t \xrightarrow{\text{Replace}} \mathcal{A}(\mathbf{x}_t)$$

$$y_t \sim p_{0t}(y_t | y); x_t \leftarrow T^{-1}[\lambda\Lambda\mathcal{P}^{-1}(\Lambda)\hat{\mathbf{y}}_t + (1 - \lambda)\Lambda T\hat{\mathbf{x}}_t + (I - \Lambda)T\hat{\mathbf{x}}_t]$$

Diffusion models for Medical Imaging inverse problem

Score-SDE , Song et al., 2022

 **Replacement-based**

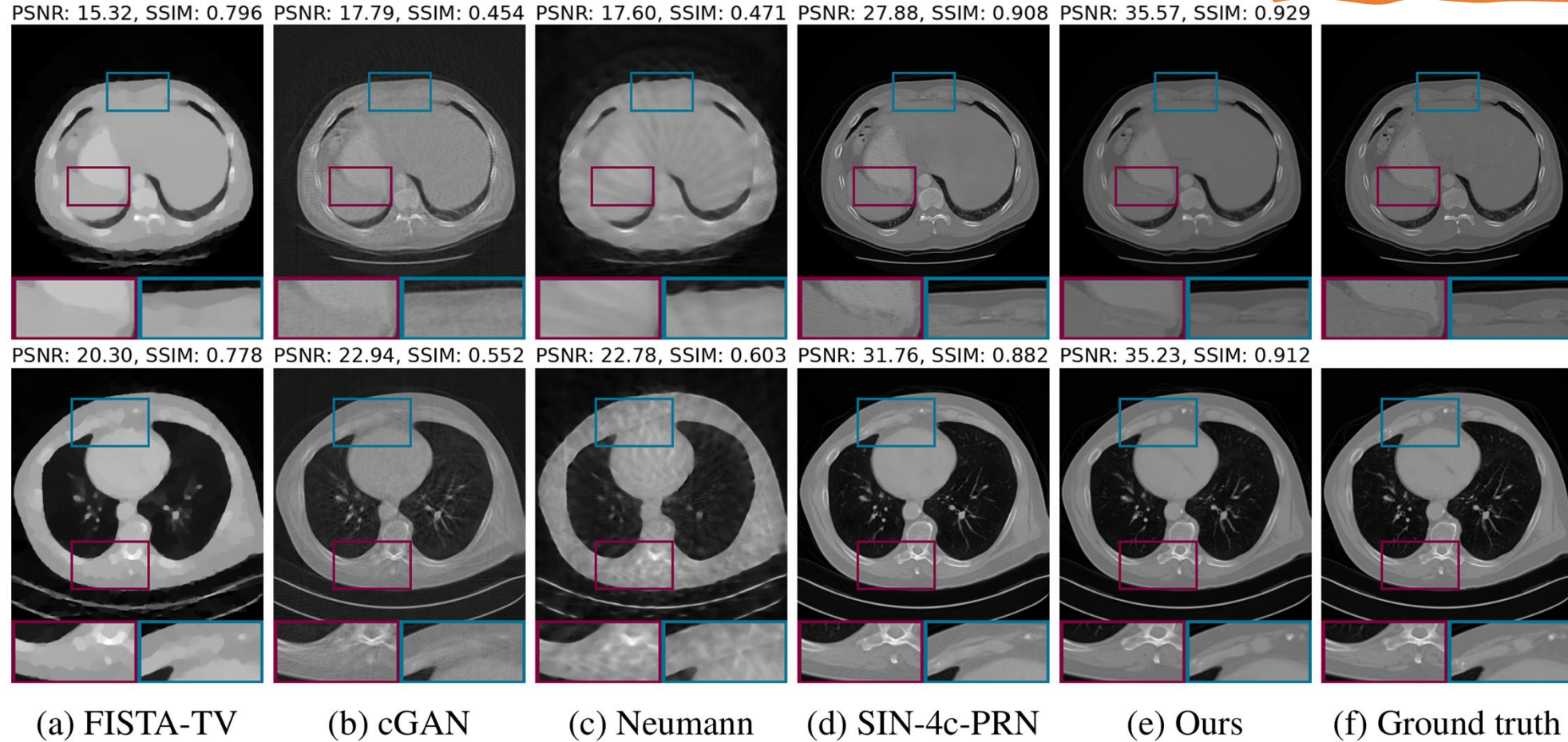
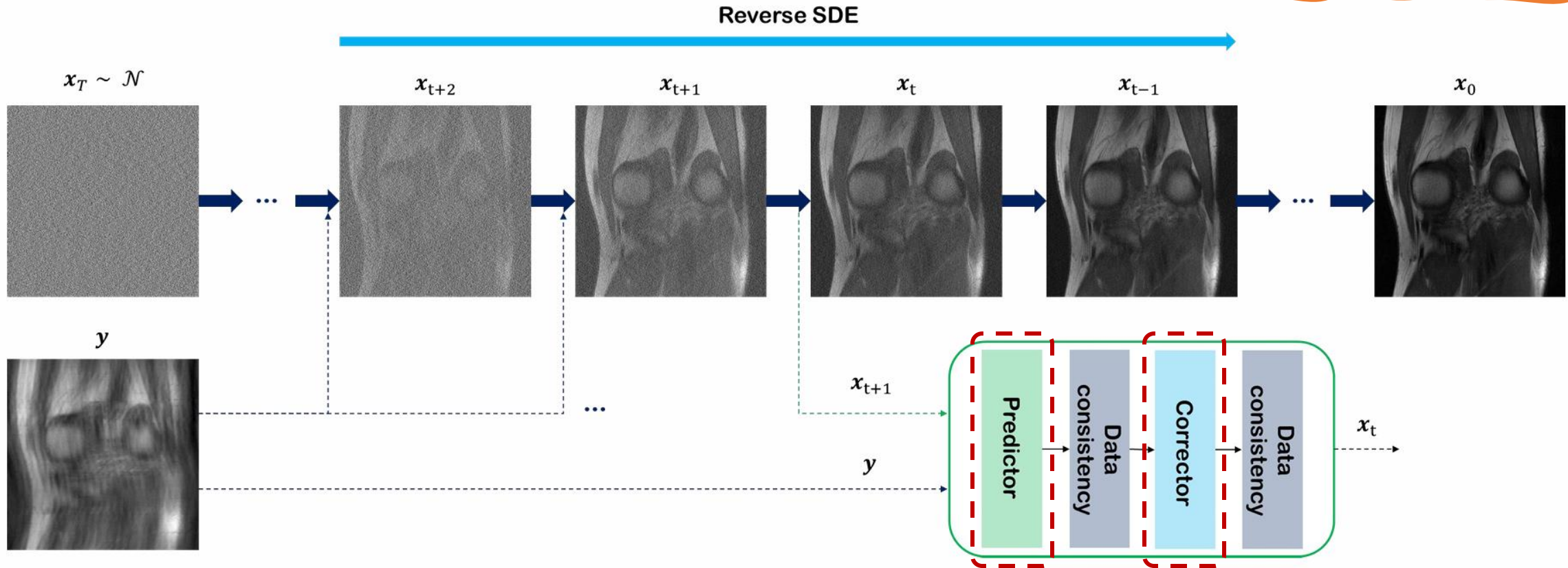


Figure 4: Examples of sparse-view CT reconstruction results on LIDC 320×320 (*Top row*) and LDCT 512×512 (*Bottom row*), all with 23 projections. You may zoom in to view more details.

Diffusion models for Medical Imaging inverse problem

Score-MRI , Chung et al., 2022

 **Replacement-based**

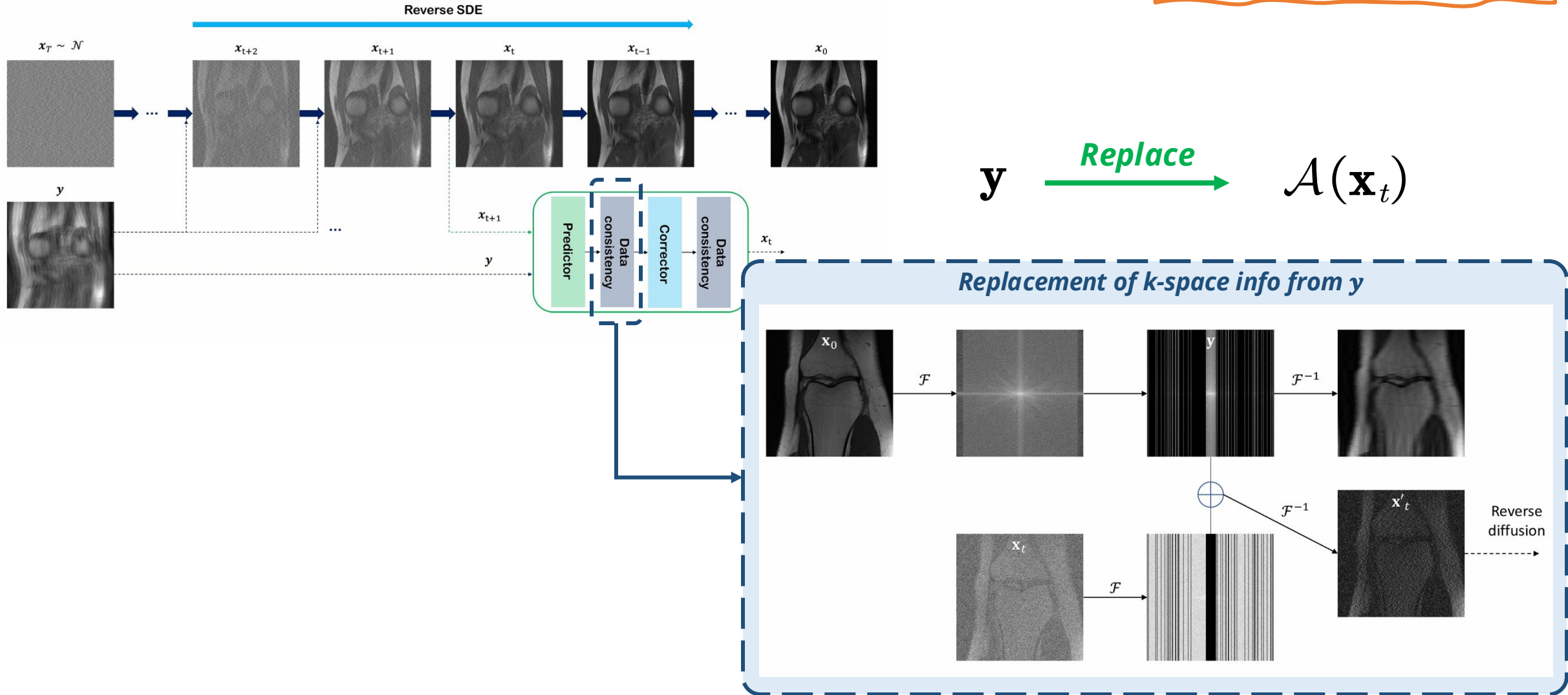


Prior: unconditional denoising

Diffusion models for Medical Imaging inverse problem

Score-MRI , Chung et al., 2022

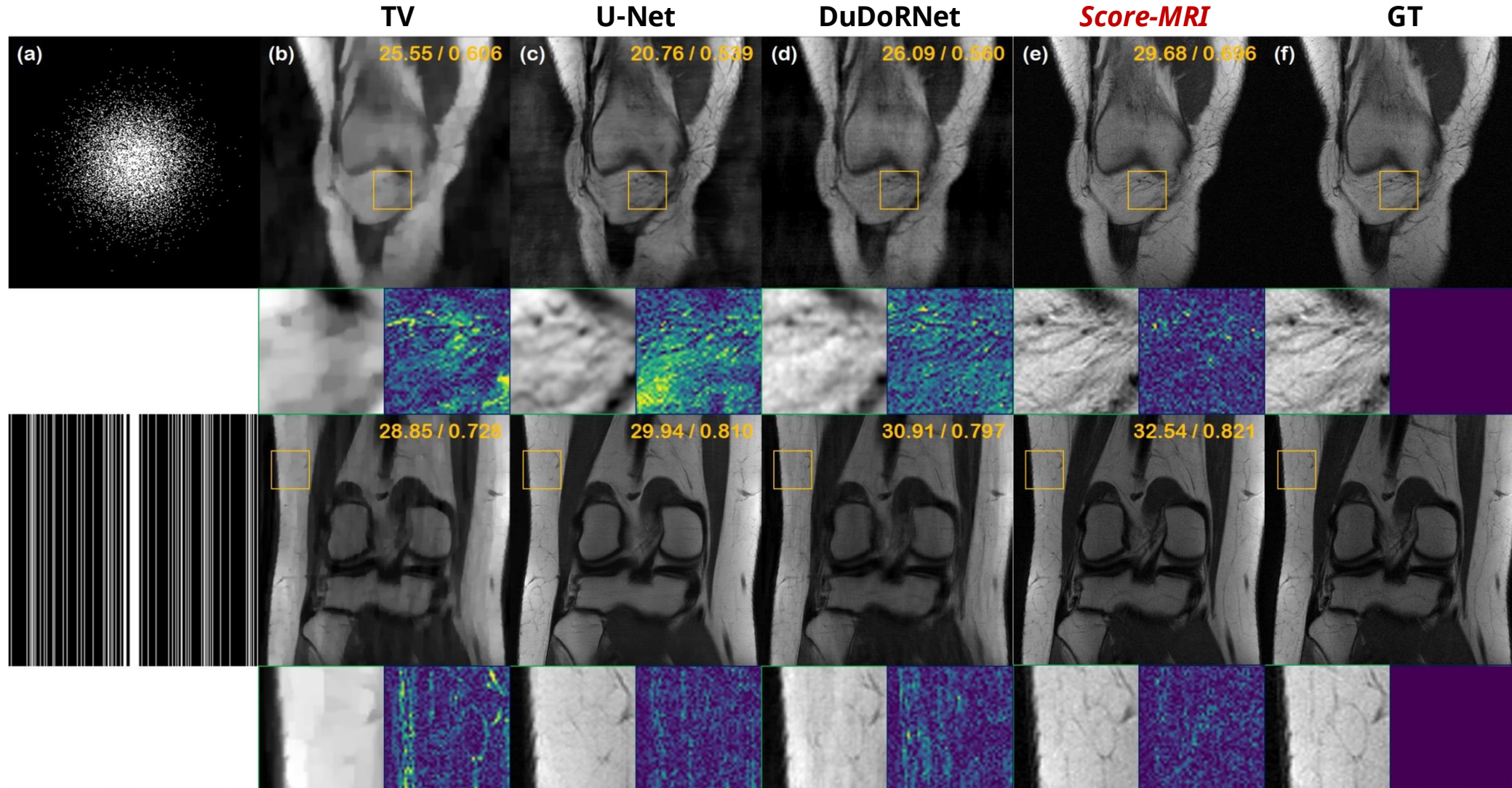
 **Replacement-based**



Diffusion models for Medical Imaging inverse problem

Score-MRI

 **Replacement-based**



Diffusion models for Medical Imaging inverse problem

DiffusionMBIR

Algorithm 1 DiffusionMBIR.

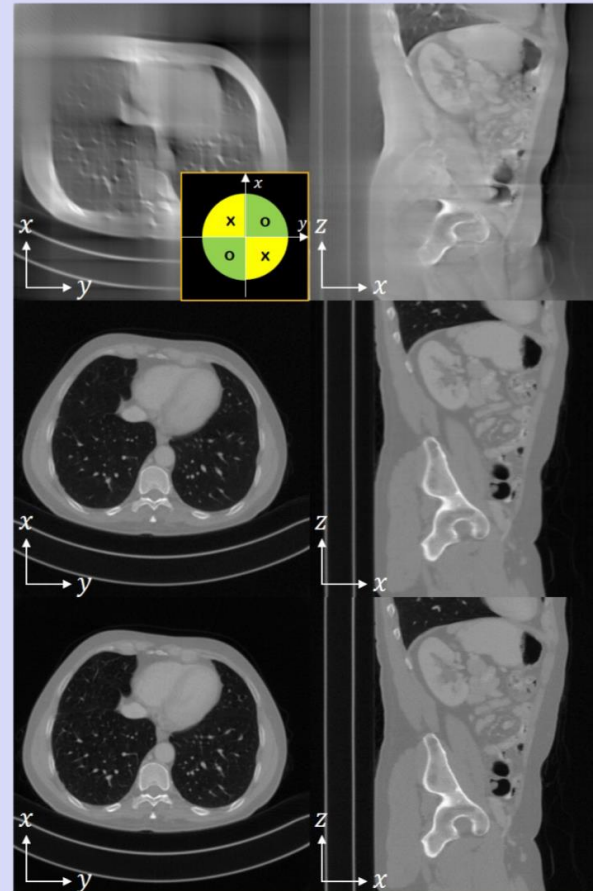
Require: Measurements $\mathbf{y}$, Pre-trained score function $\mathbf{s}_{\theta^*}$, Timesteps T , Noise schedule $\{\sigma_t\}_{t=0}^T$.

- 1: $\mathbf{x}_T \sim (\mathbf{0}, \sigma_T^2 \mathbf{I})$;
- 2: **for** $t = T - 1, \dots, 0$ **do**
- 3: $\triangleright$ *1. Distribution Prior Sub-problem*
- 4: $\mathbf{x}_t = \text{Sampler}(\mathbf{s}_{\theta^*}, \mathbf{x}_{t+1}, t + 1)$
- 5: $\triangleright$ *2. Data Fidelity Sub-problem*
- 6: $\mathbf{x}_t = \text{CG}(\mathbf{A}^T \mathbf{A}, \mathbf{A}^T \mathbf{y}, \mathbf{x}_{t+1})$
- 7: **end for**
- 8: **return** $\mathbf{x}_0$

Reconstruction-based

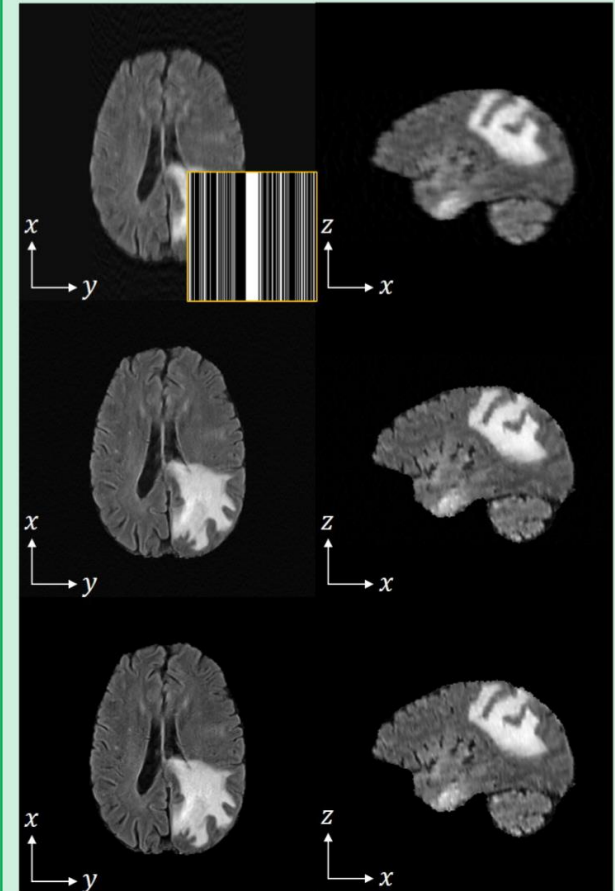
Limited-angle tomography

In-distribution



Compressed-sensing MRI

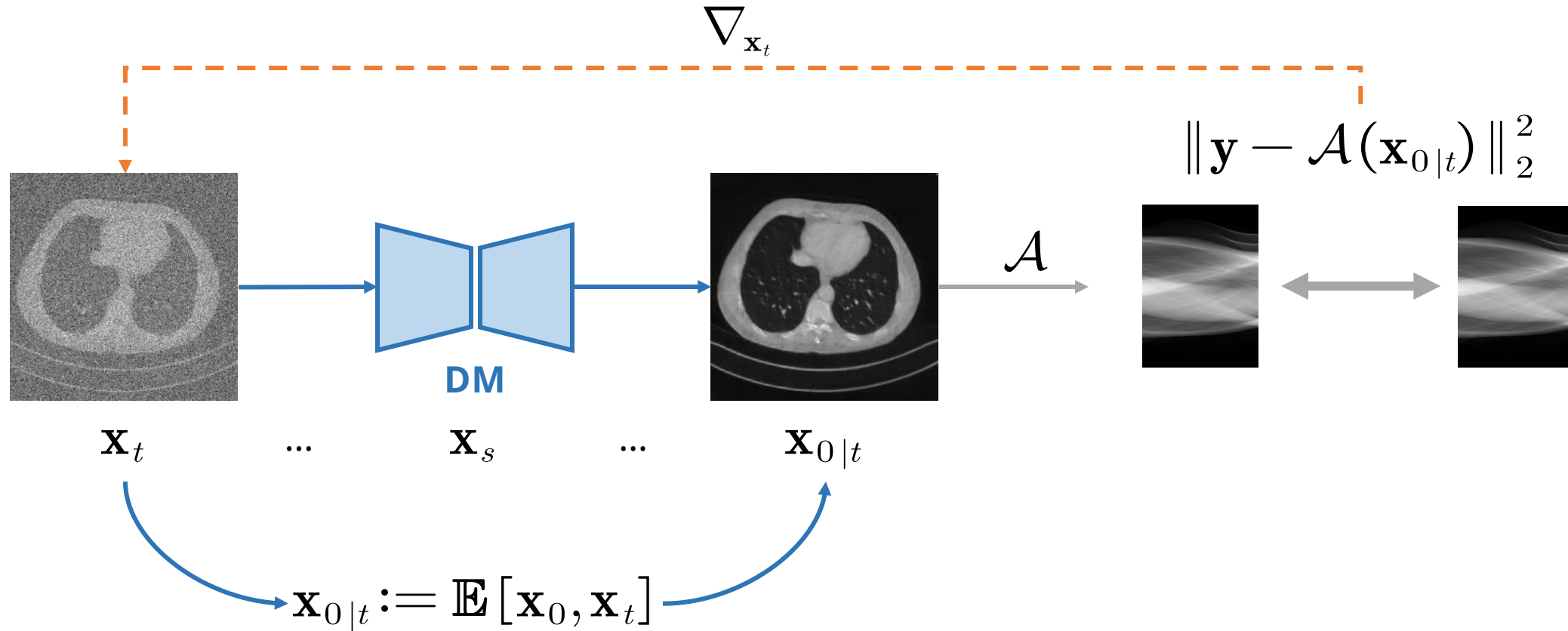
Out-of-distribution



Diffusion models for Medical Imaging inverse problem

DPS

Reconstruction-based



$$\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t, \mathbf{y}] \approx \mathbb{E}[\mathbf{x}_0, \mathbf{x}_t] + \frac{t^2}{2\sigma_y^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathcal{A}(\mathbb{E}[\mathbf{x}_0, \mathbf{x}_t])\|_2^2$$

Medical imaging acquisition:

Reconstruction-based

$$\mathbf{y} = \mathcal{A}\mathbf{x} + \mathbf{n}$$

Measurement data Forward model Desired image System Noise

$\mathbf{y} \in \mathbb{R}^m$ $\mathcal{A} \in \mathbb{R}^{m \times n}$ $\mathbf{x} \in \mathbb{R}^n$ $\mathbf{n} \in \mathbb{R}^m$

Reconstruction

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \underbrace{\|\mathbf{y} - \mathcal{A}(\mathbf{x})\|_2^2}_{\text{Data Fidelity term}} + \lambda \cdot \underbrace{\mathcal{R}(\mathbf{x})}_{\text{Prior term}}$$

Decompose the original inverse problem

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \frac{1}{2} \|\mathbf{y} - \mathcal{A}(\mathbf{x})\|_2^2 + \lambda \cdot \mathcal{R}(\mathbf{x})$$



HQS algorithm



$$\begin{cases} \mathbf{x} \leftarrow \text{prox}_{\|\cdot\|_2, \mu}(\mathbf{z}) = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{Ax}\|_2^2 + \mu \|\mathbf{x} - \mathbf{z}\|^2 & \text{Solve Fidelity sub-Problem} \\ \mathbf{z} \leftarrow \text{prox}_{\mathcal{R}, \lambda}(\mathbf{x}) = \arg \min_{\mathbf{z}} \frac{\mu}{2} \|\mathbf{z} - \mathbf{x}\|_2^2 + \lambda \cdot \mathcal{R}(\mathbf{z}) & \text{Solve Prior sub-Problem} \end{cases}$$

Plug-and-Play method

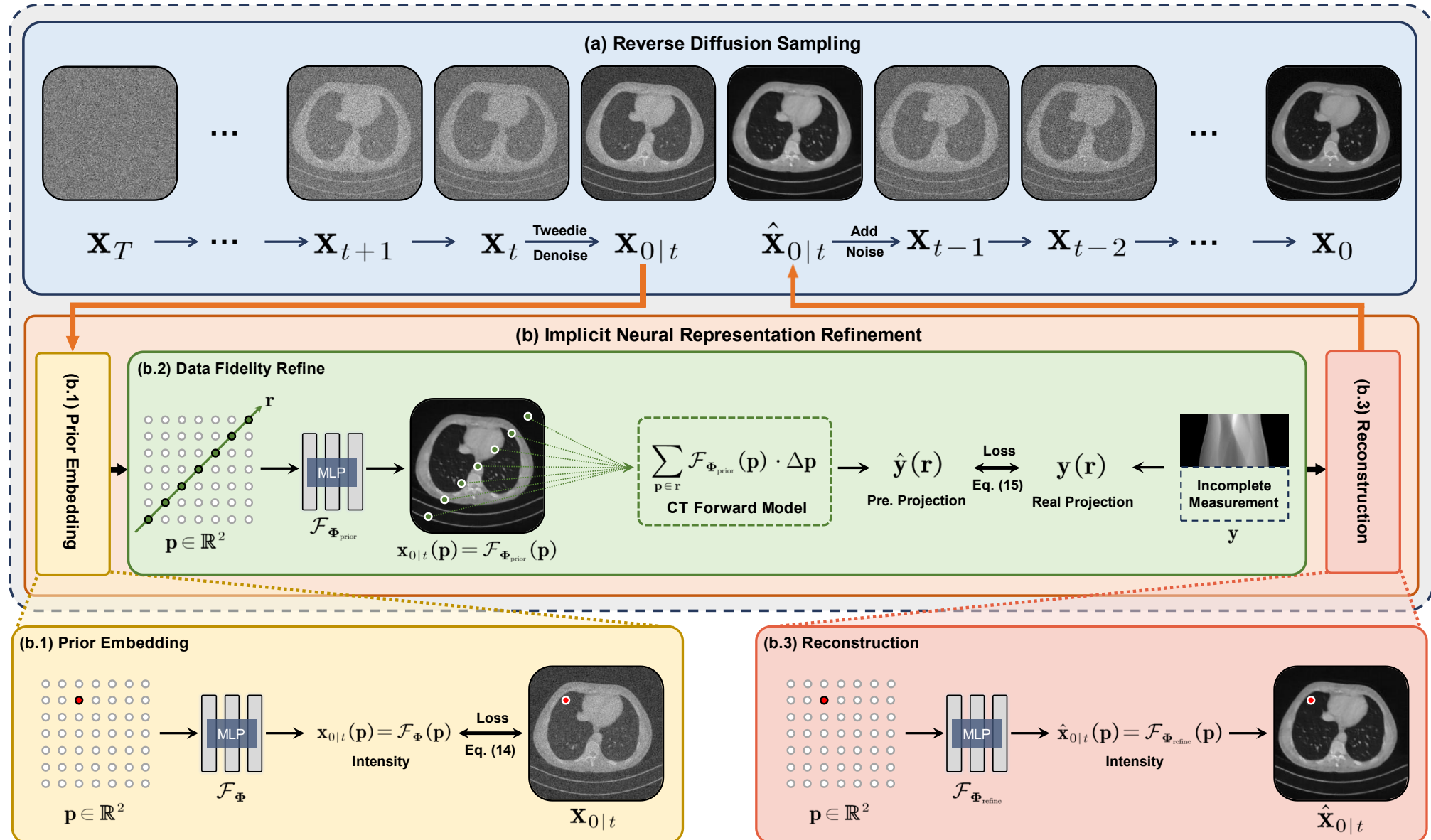
$$\begin{cases} \mathbf{x} \leftarrow \text{prox}_{\|\cdot\|_2, \mu}(\mathbf{z}) = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{Ax}\|_2^2 + \mu \|\mathbf{x} - \mathbf{z}\|^2 & \text{Solve Fidelity sub-Problem} \\ \mathbf{z} \leftarrow \text{prox}_{\mathcal{R}, \lambda}(\mathbf{x}) = \arg \min_{\mathbf{z}} \frac{\mu}{2} \|\mathbf{z} - \mathbf{x}\|_2^2 + \lambda \cdot \mathcal{R}(\mathbf{z}) & \text{Solve Prior sub-Problem} \end{cases}$$

$$\begin{cases} \mathbf{z}_t = \mathbf{x}_{0|t} \approx \mathbf{x}_t + \sigma_t^2 \mathbf{s}_{\theta^*}(\mathbf{x}_t, t), \\ \mathbf{x}_{t-1} = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{Ax}\|^2 + \mu \sigma_i^2 \|\mathbf{x} - \mathbf{x}_{0|t}\|^2 + \sigma_{t-1} \epsilon \end{cases}$$

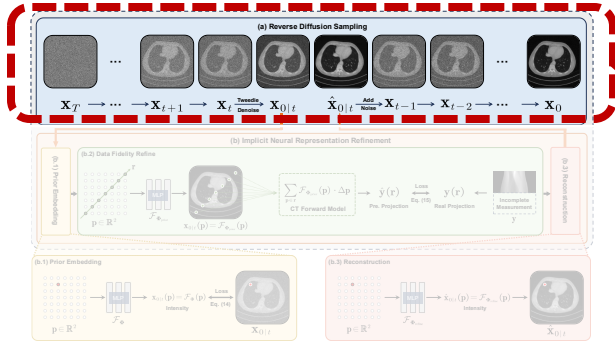
Diffusion denoising

INR-based fidelity refinement

DPER: Framework



DPER: Framework

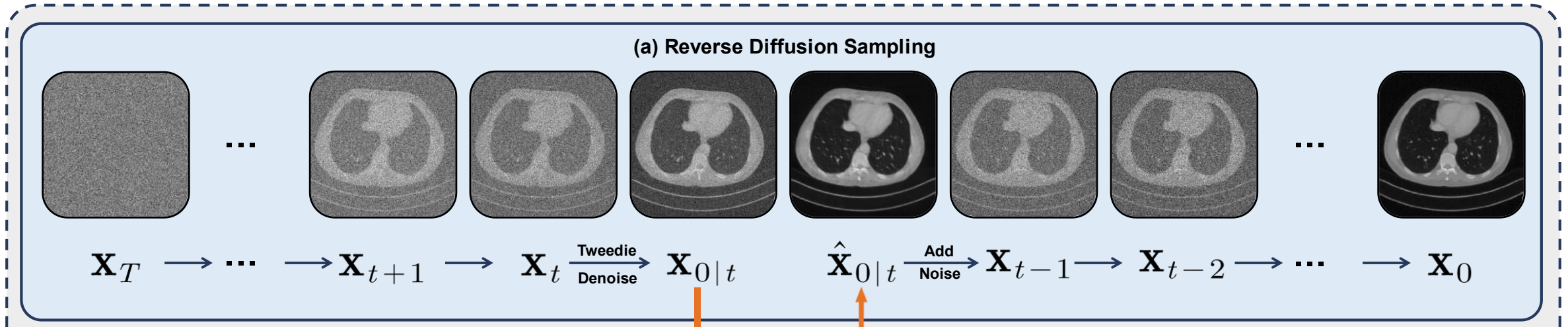


Solve Prior sub-Problem

$$\mathbf{z} \leftarrow \text{prox}_{\mathcal{R}, \lambda}(\mathbf{x}) = \arg \min_{\mathbf{z}} \frac{\mu}{2} \|\mathbf{z} - \mathbf{x}\|_2^2 + \lambda \cdot \mathcal{R}(\mathbf{z})$$

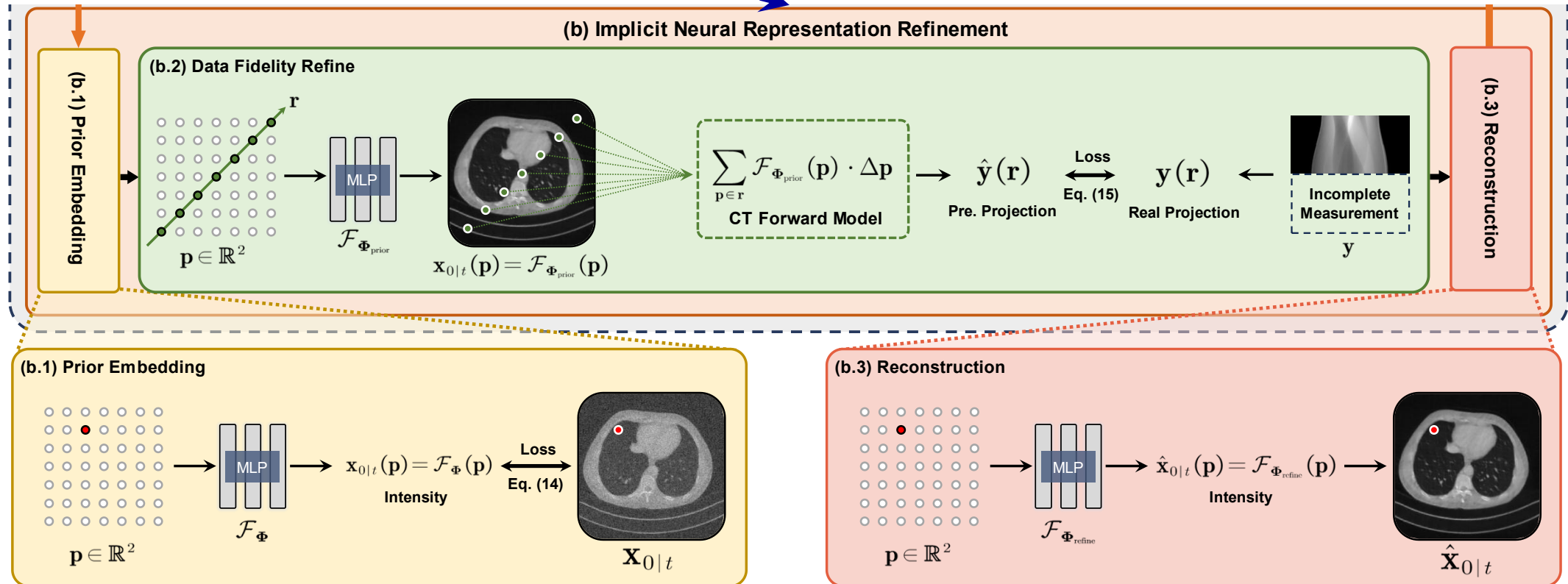
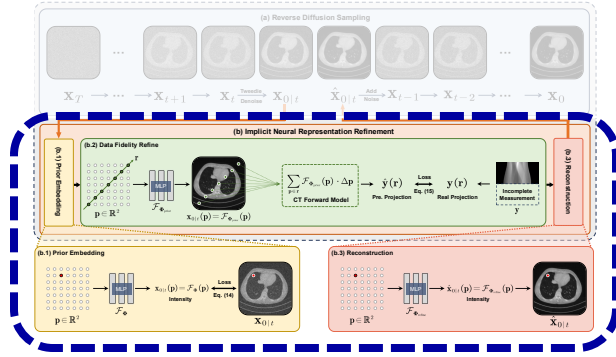
Diffusion denoising process

$$\mathbf{z}_t = \mathbf{x}_{0|t} \approx \mathbf{x}_t + \sigma_t^2 \mathbf{s}_{\theta^*}(\mathbf{x}_t, t)$$



Solve Fidelity sub-Problem

$$\mathbf{x} \leftarrow \text{prox}_{\|\cdot\|_2, \mu}(\mathbf{z}) = \arg \min_{\mathbf{x}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \mu \|\mathbf{x} - \mathbf{z}\|^2$$

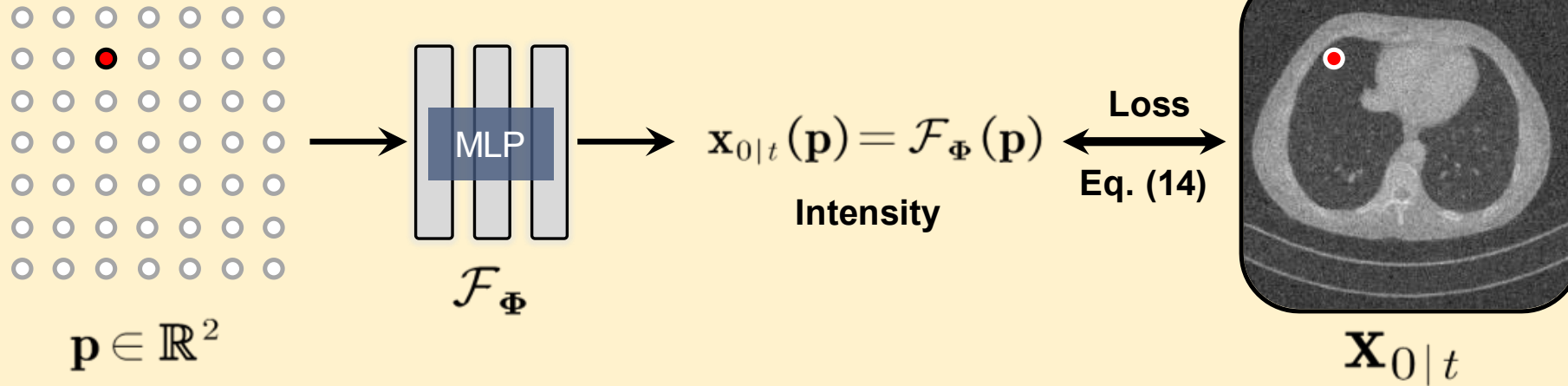


INR-based Fidelity Refinement:

1. Prior embedding

$$\Phi_{\text{prior}} = \underset{\Phi}{\operatorname{argmin}} \frac{1}{N} \sum_{i=1}^N \|\mathcal{F}_{\Phi}(\mathbf{p}_i) - \mathbf{x}_{0|t}(\mathbf{p}_i)\|^2$$

(b.1) Prior Embedding



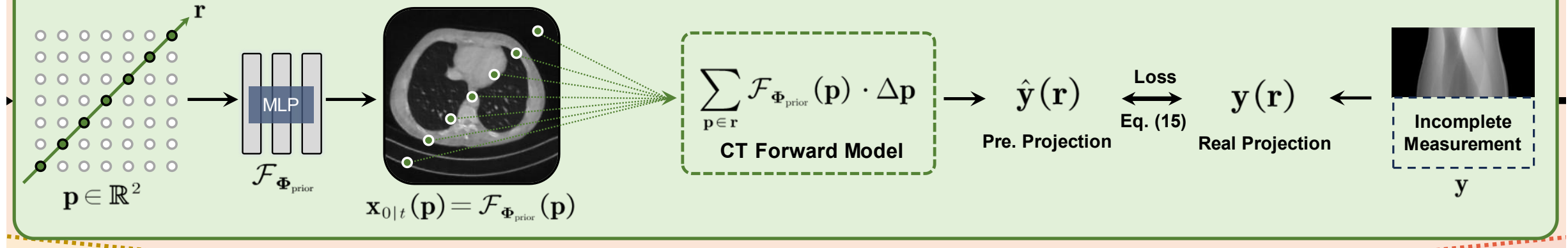
INR-based Fidelity Refinement:

2. Data Fidelity Refine

$$\Phi_{\text{refine}} = \underset{\Phi_{\text{prior}}}{\operatorname{argmin}} \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} |\hat{\mathbf{y}}(\mathbf{r}) - \mathbf{y}(\mathbf{r})|,$$

$$\hat{\mathbf{y}}(\mathbf{r}) = \sum_{\mathbf{p} \in \mathbf{r}} \mathcal{F}_{\Phi_{\text{prior}}}(\mathbf{p}) \cdot \Delta \mathbf{p}$$

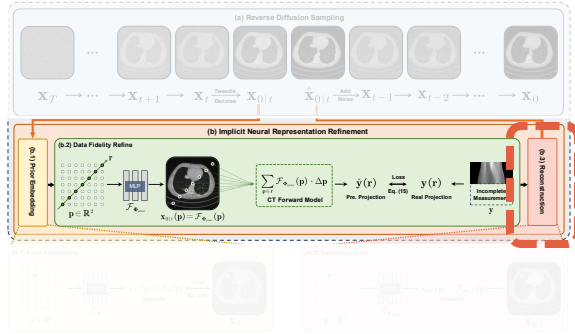
(b.2) Data Fidelity Refine



DPER: Framework

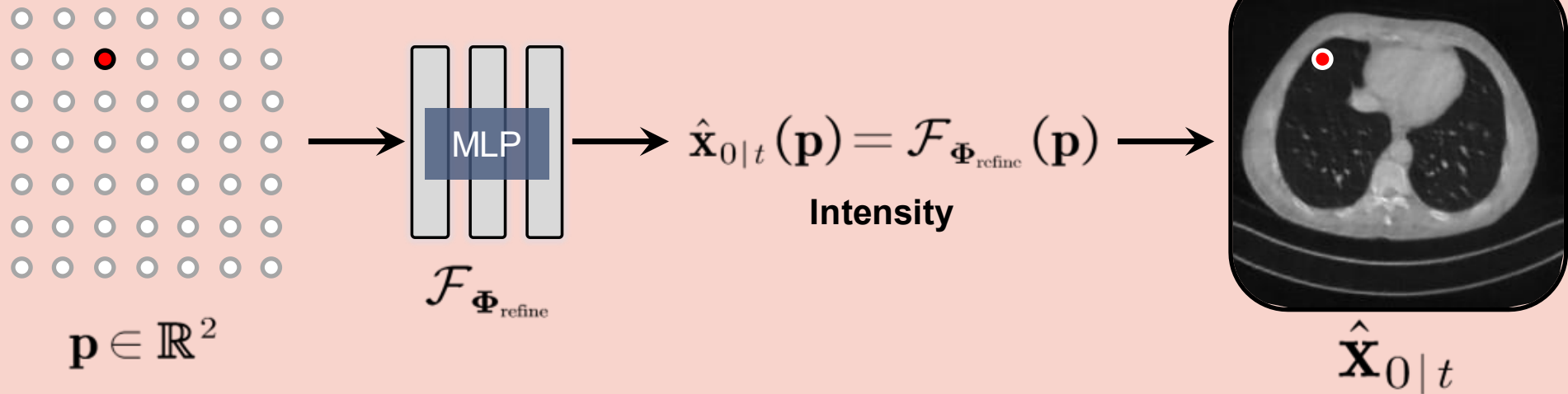
INR-based Fidelity Refinement:

3. Reconstruction

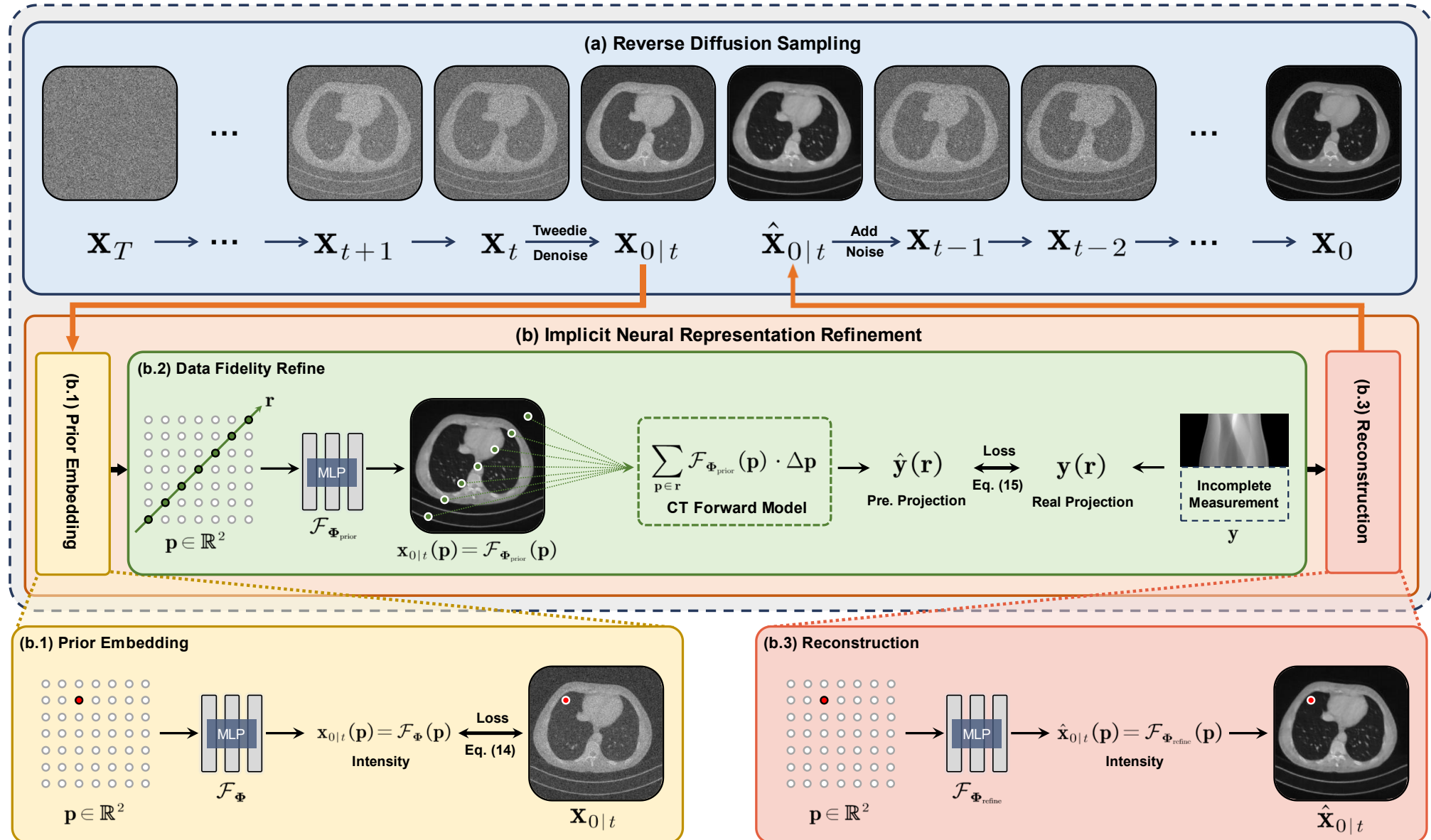


$$\hat{\mathbf{X}}_{0|t} = \mathcal{F}_{\Phi_{\text{refine}}}(\mathbf{p})$$

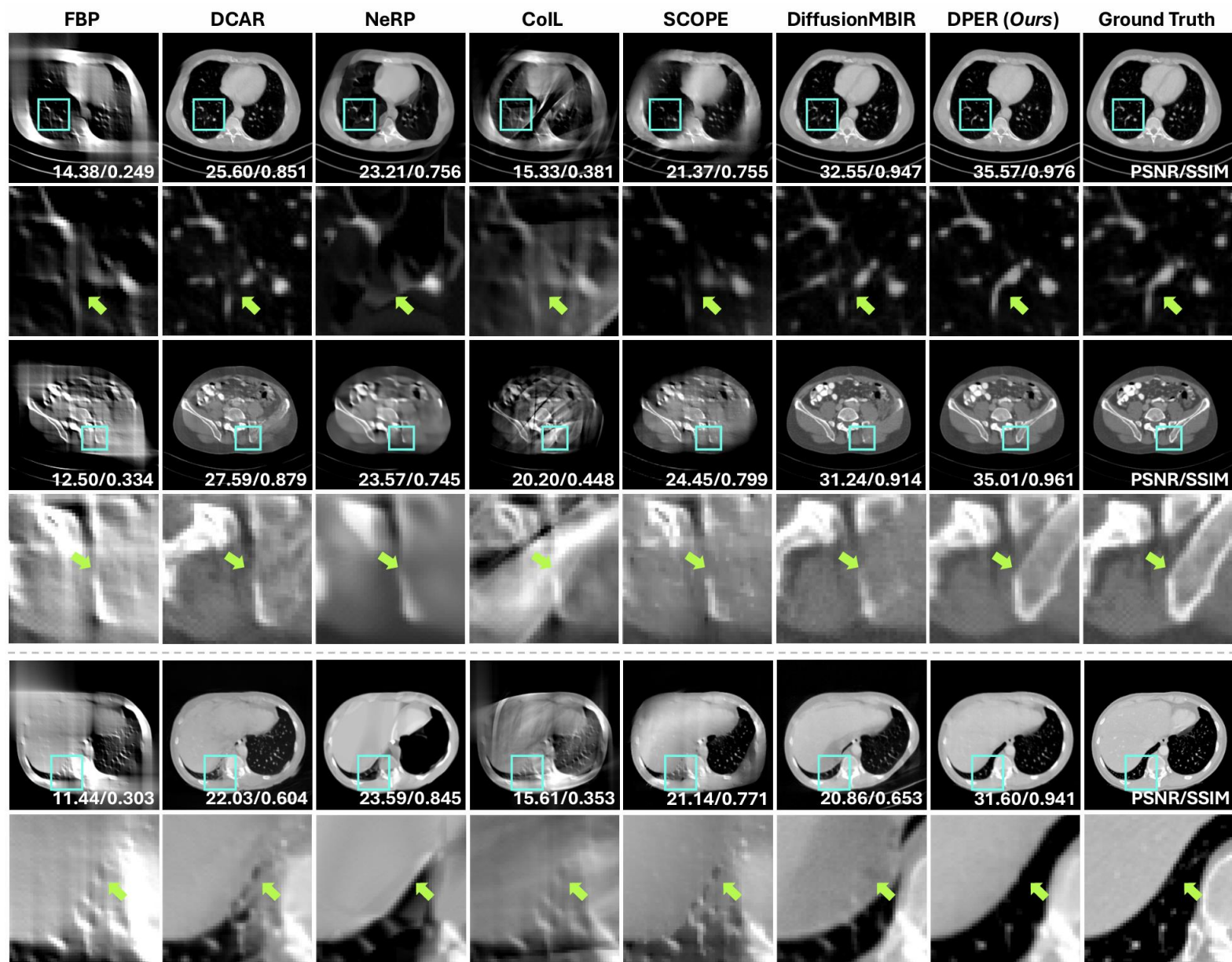
(b.3) Reconstruction



DPER: Framework

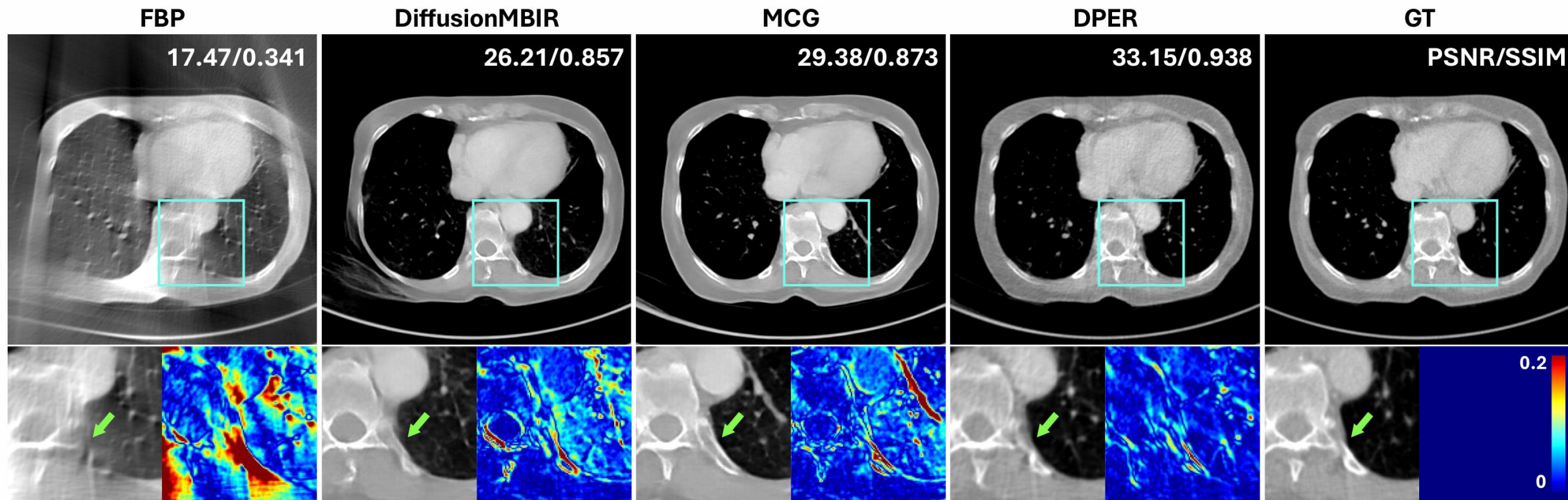


DPER: Framework



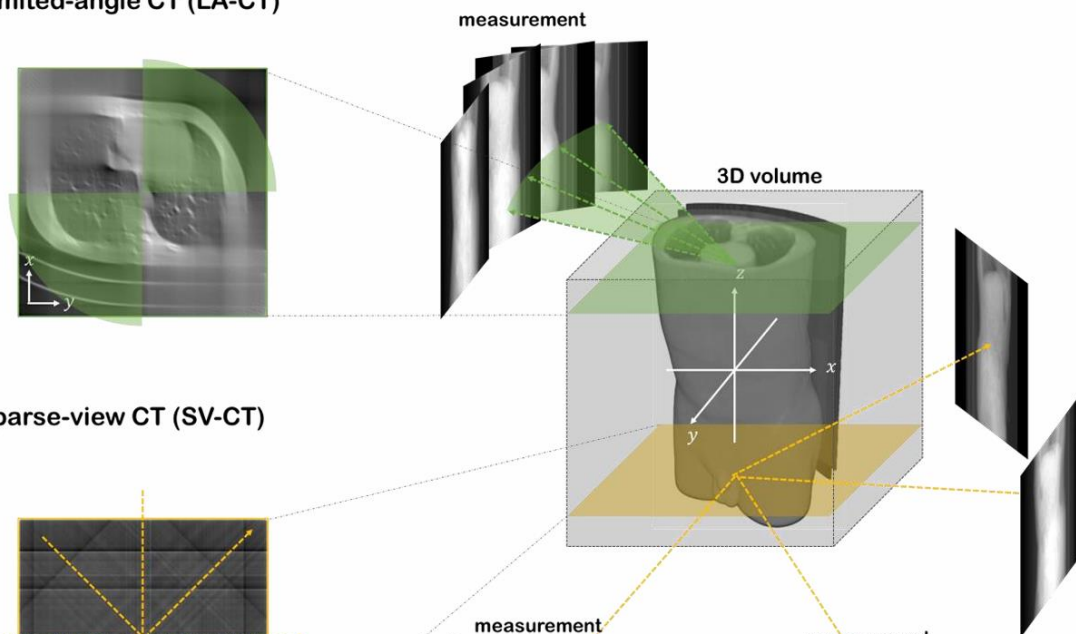
DPER: Framework

LACT-[0, 120]°

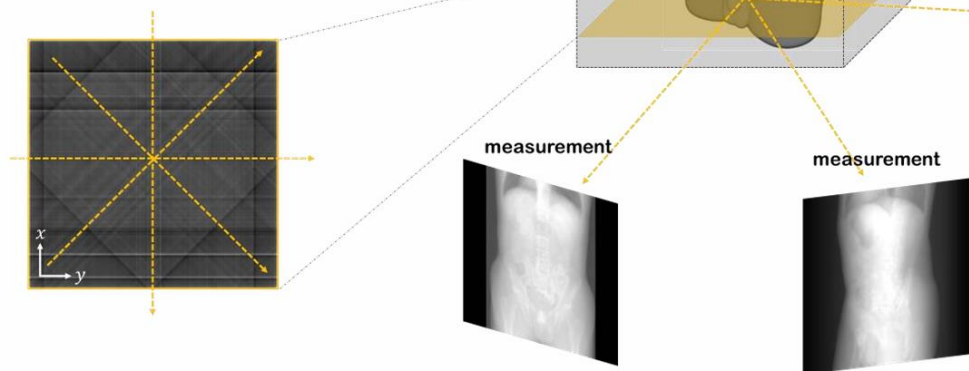


| 3D Problem

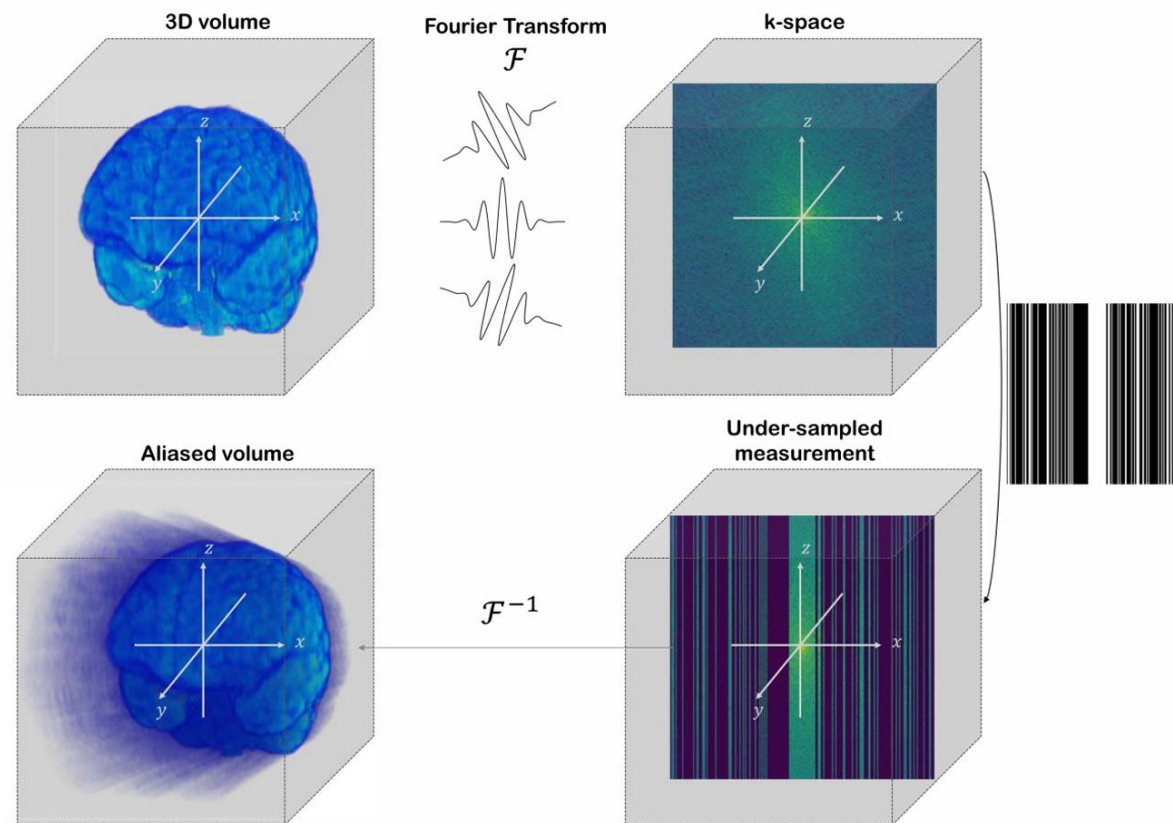
(a) Limited-angle CT (LA-CT)



(b) Sparse-view CT (SV-CT)



(c) Compressed-sensing MRI



| 3D Problem

Challenges to solving the 3D problem

- 3D representations are **memory heavy**
 - **Voxels**: Hard to deal with $> 128^3$ data
 - **Point clouds**: Sparse representation, but not suitable for medical imaging inverse problems
- **3D voxel diffusion?**
 - The whole diffusion process **stays in the data dimension**
 - **Computationally too heavy**

| 3D Problem

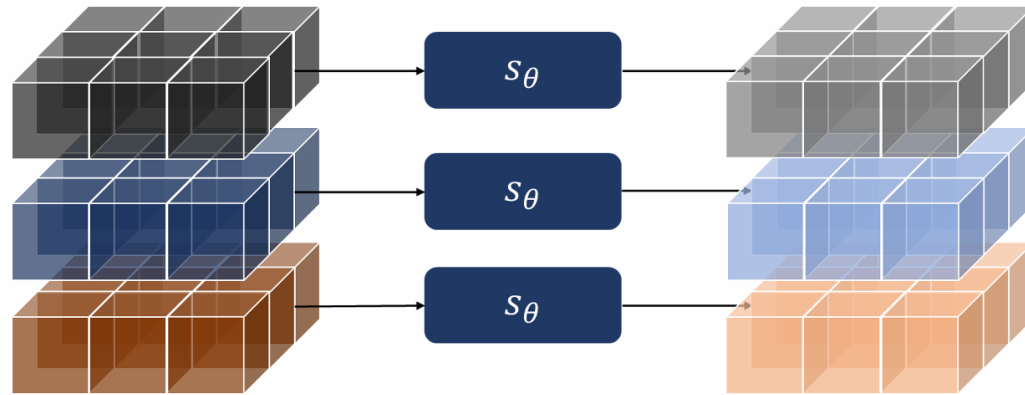
One Solution...

Solving **3D** problem using **2D**
diffusion prior

3D Problem: Add TV loss in the z-axis

DiffusionMBIR & DDS

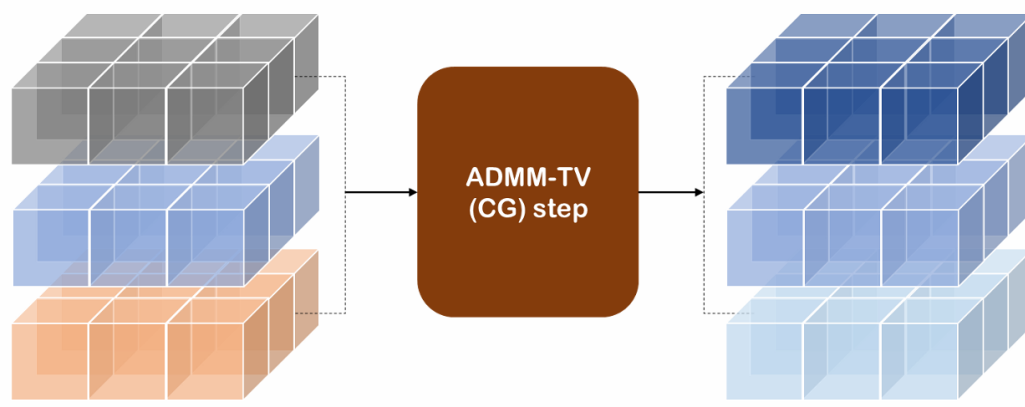
1. Score function denoising (parallel)



1. Denoising with score function (**parallel**)

$$\mathbf{x}'_{i-1} \leftarrow (\sigma_i^2 - \sigma_{i-1}^2) \mathbf{s}_{\theta^*}(\mathbf{x}_t, t) + \sqrt{\sigma_i^2 - \sigma_{i-1}^2} \epsilon$$

2. ADMM-TV (joint)



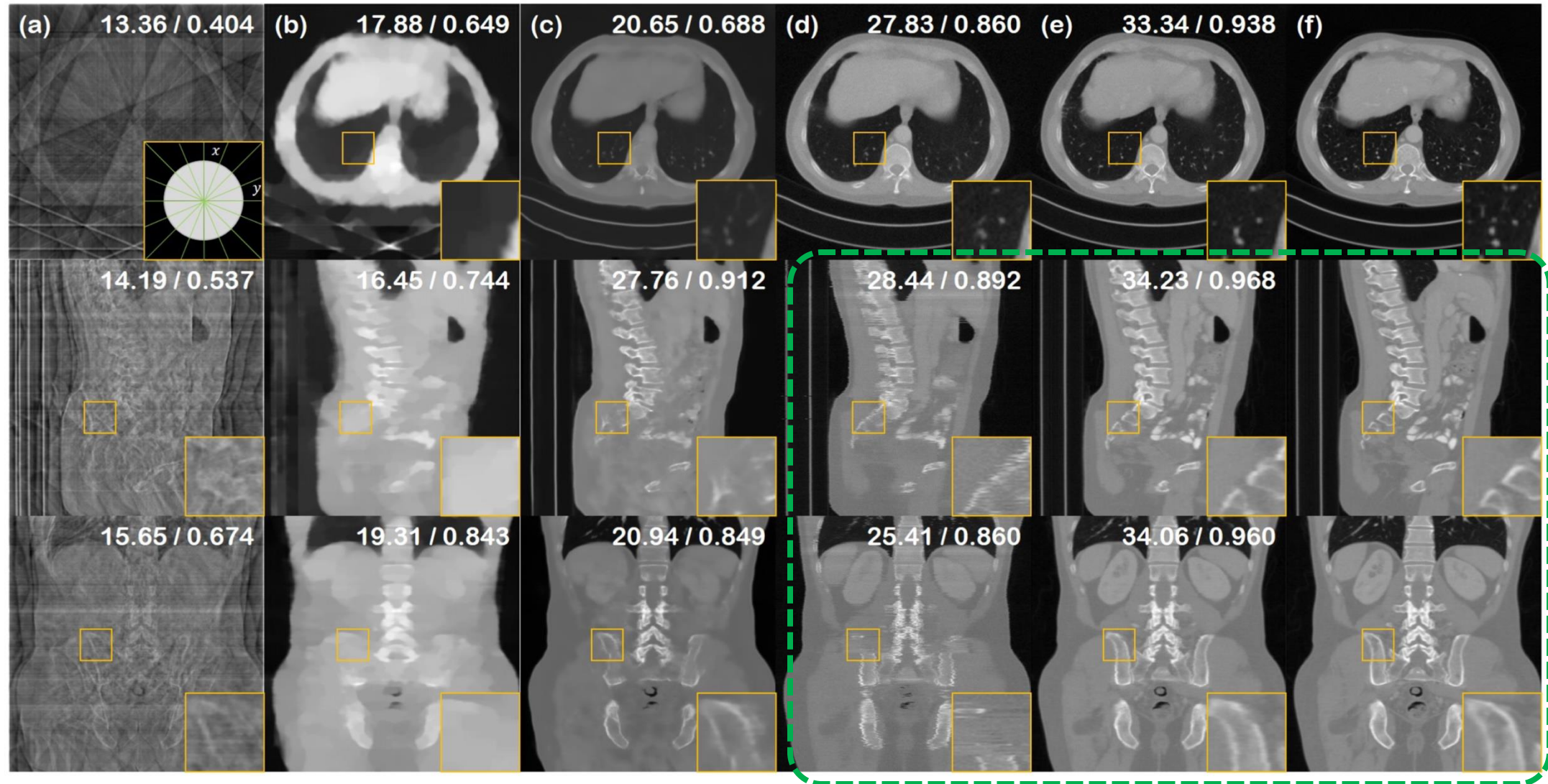
2. Data consistency + TV prior augmenting (**joint**)

$$\mathbf{x}_{i-1} \leftarrow \underset{\mathbf{x}'_{i-1}}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}'_{i-1}\|_2^2 + \underbrace{\|\mathbf{D}_z \mathbf{x}'_{i-1}\|_1}_{\text{TV Loss}}$$

Effectively solved with, e.g. ADMM

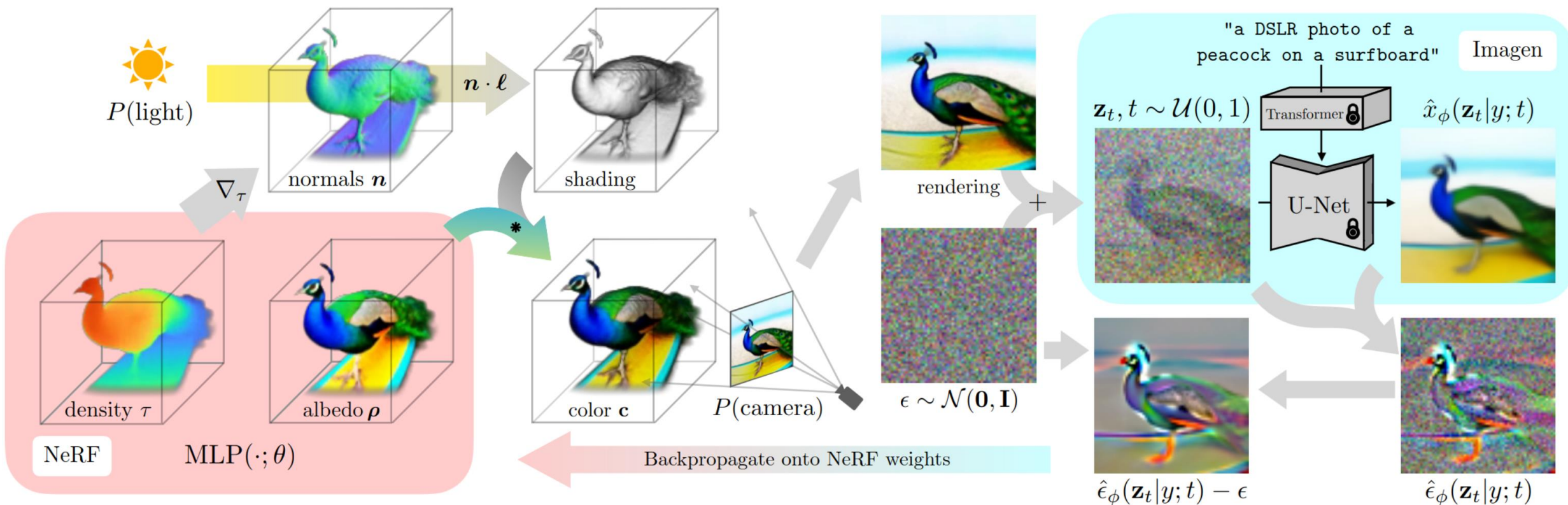
3D Problem: Add TV loss in the z-axis

DiffusionMBIR & DDS



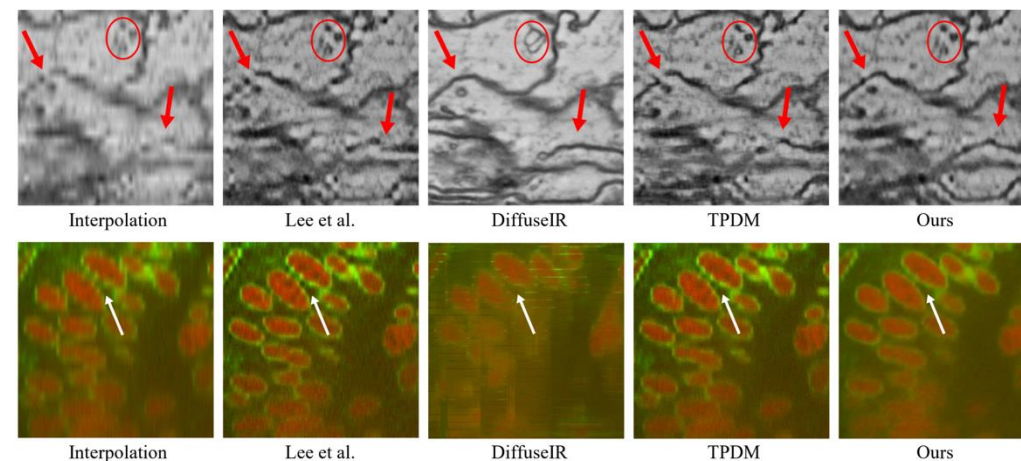
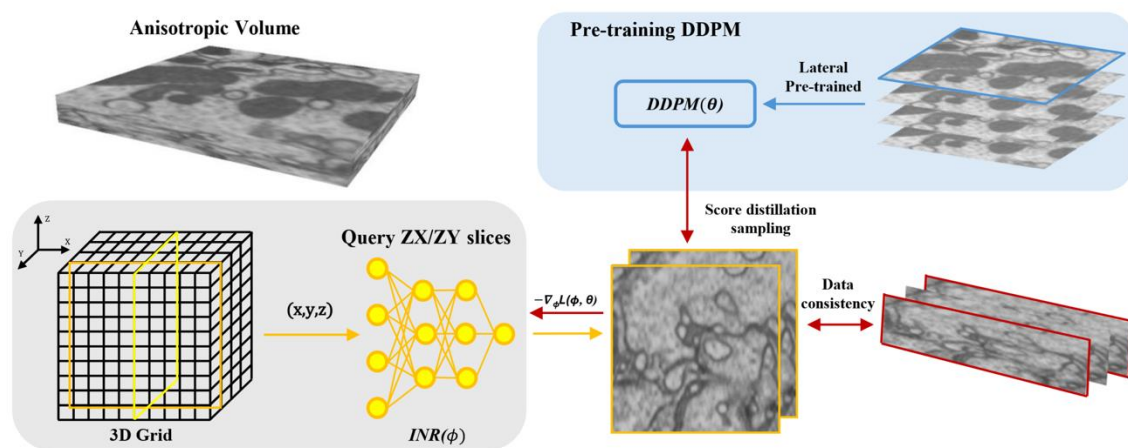
3D Problem: *DreamFusion-Style*

DreamFusion-Style



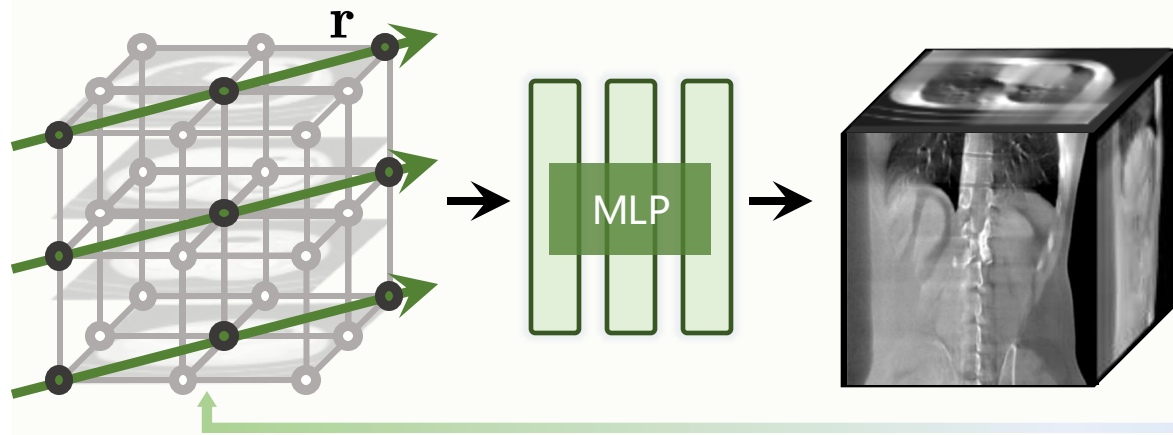
3D Problem: *DreamFusion-Style*

(a) 3D microscopy SR

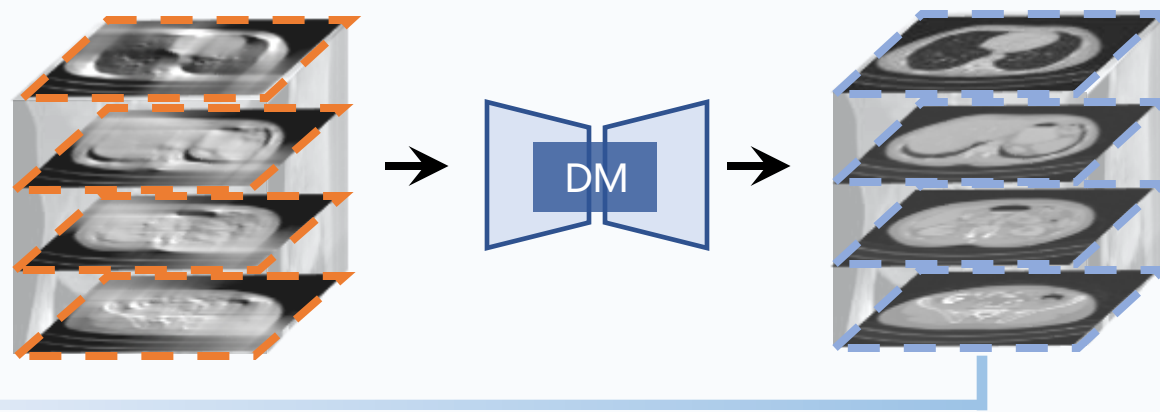


(b) 3D DPFR

(1) INR-based CT recon



(2) Diffusion one-step regularization



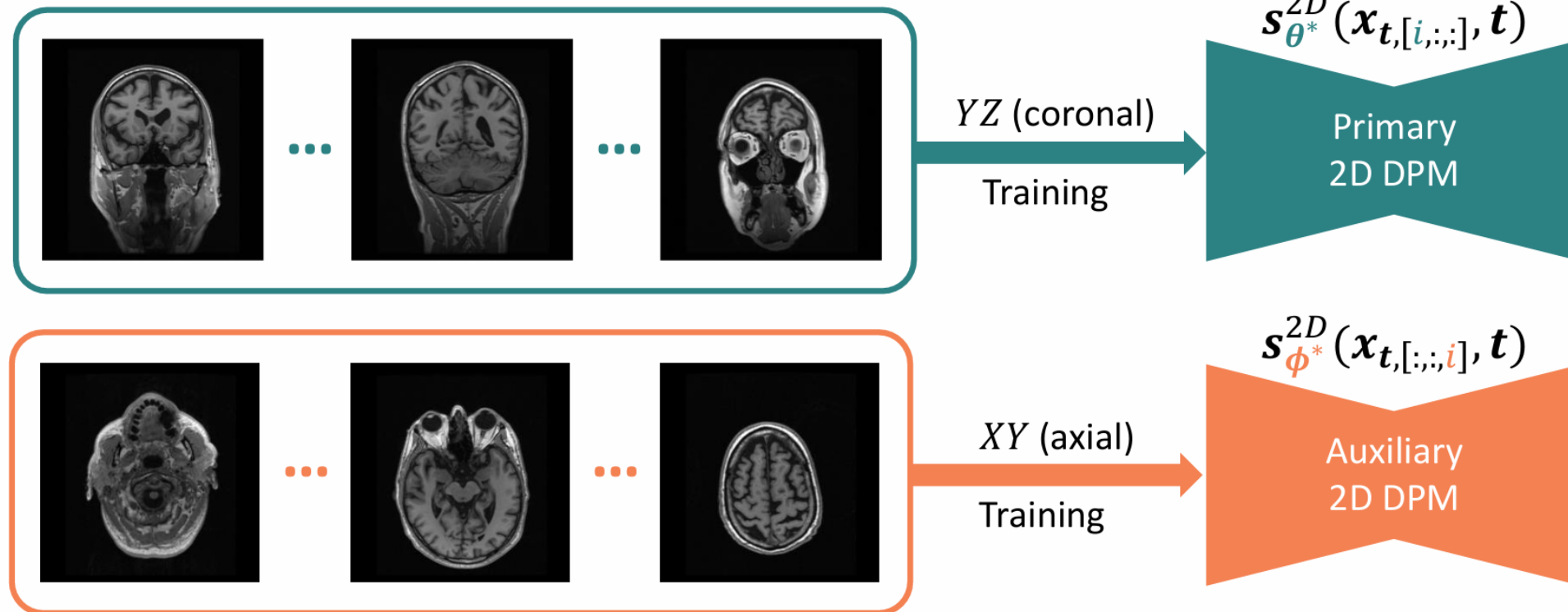
Fitting & Backpropagate

3D Problem: *multiple 2D diffusion*

TPDM (Lee et al., 2023)

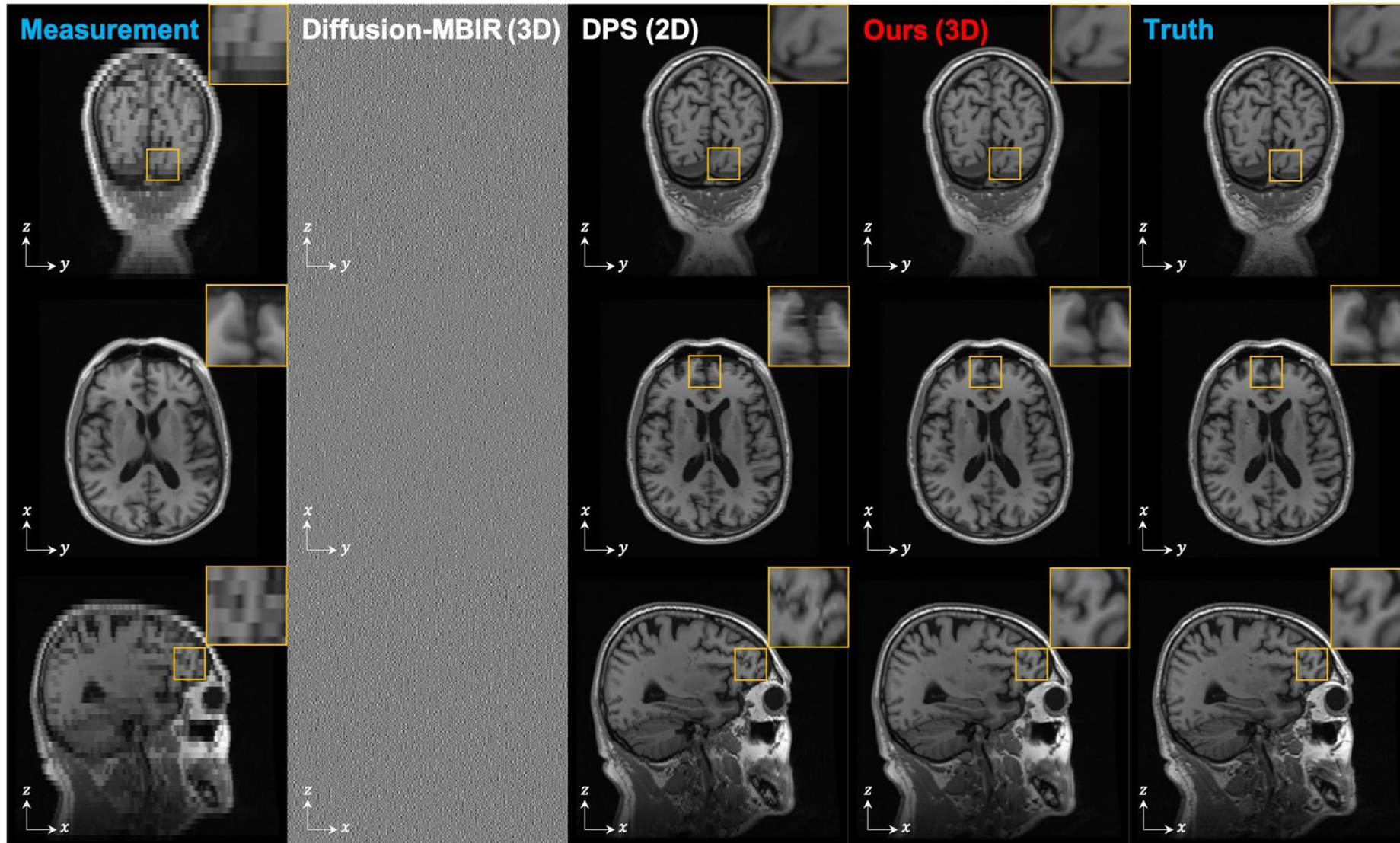
Let's use **two** 2D diffusion models trained in **different plane orientations**!

Two Perpendicular Diffusion Models (TPDM)



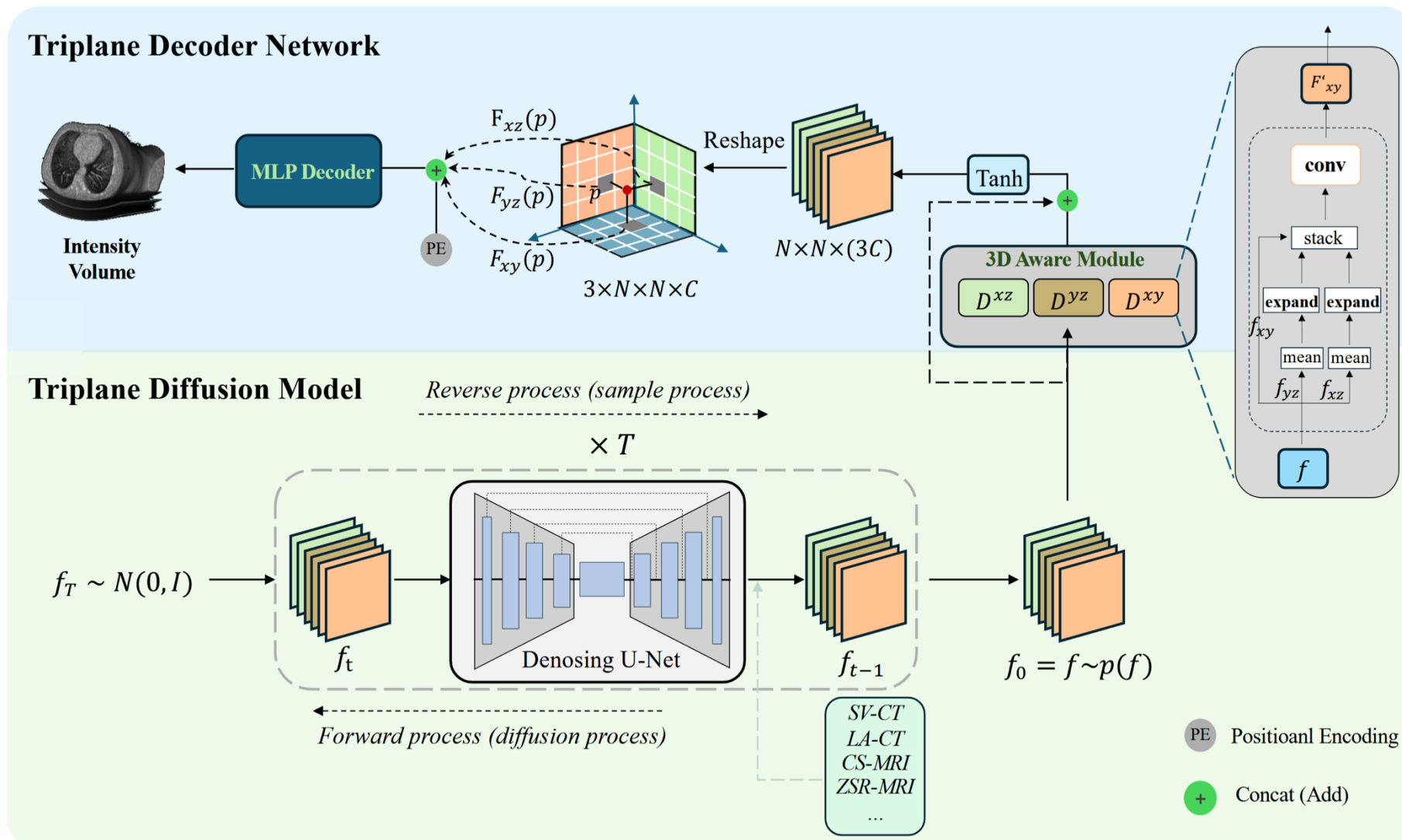
3D Problem: *multiple 2D diffusion*

TPDM (Lee et al., 2023)



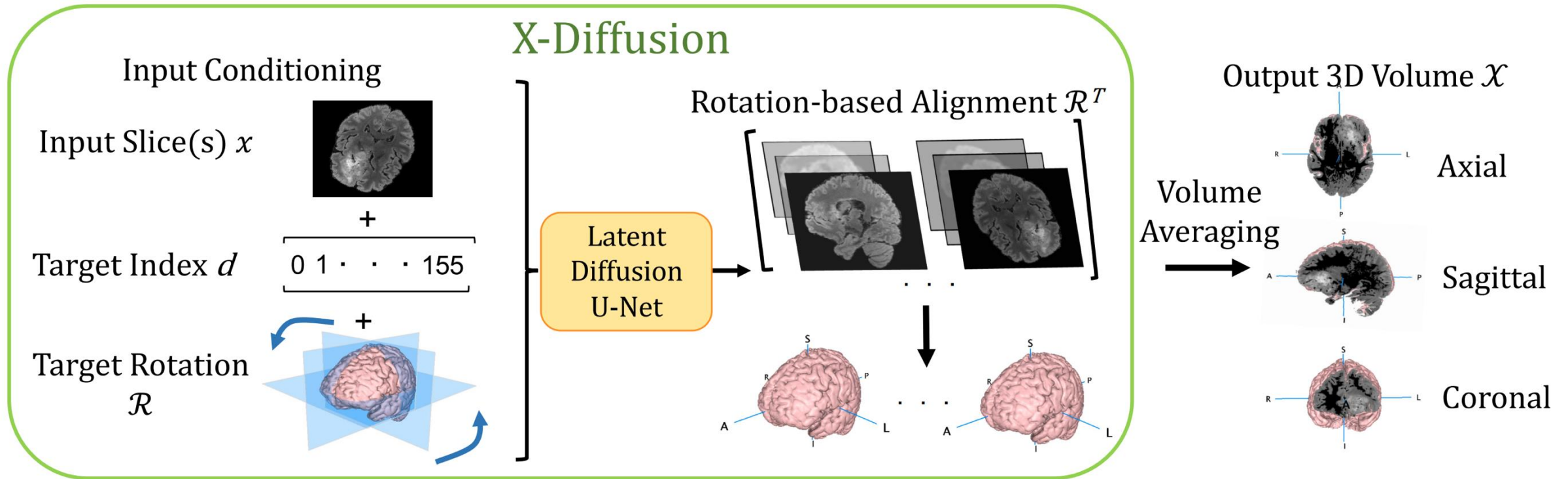
3D Problem : 3D generative model

Blaze3DM



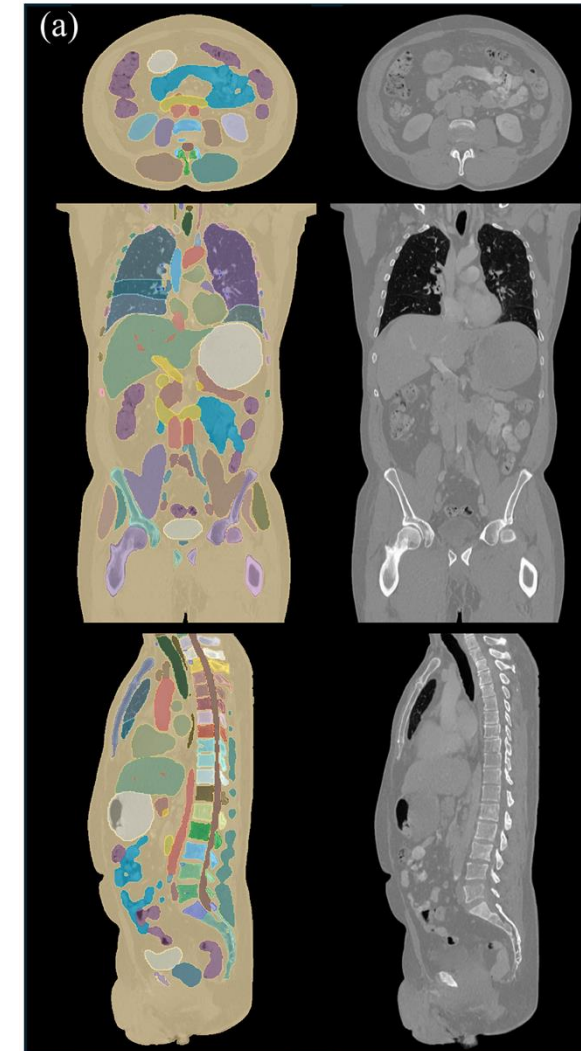
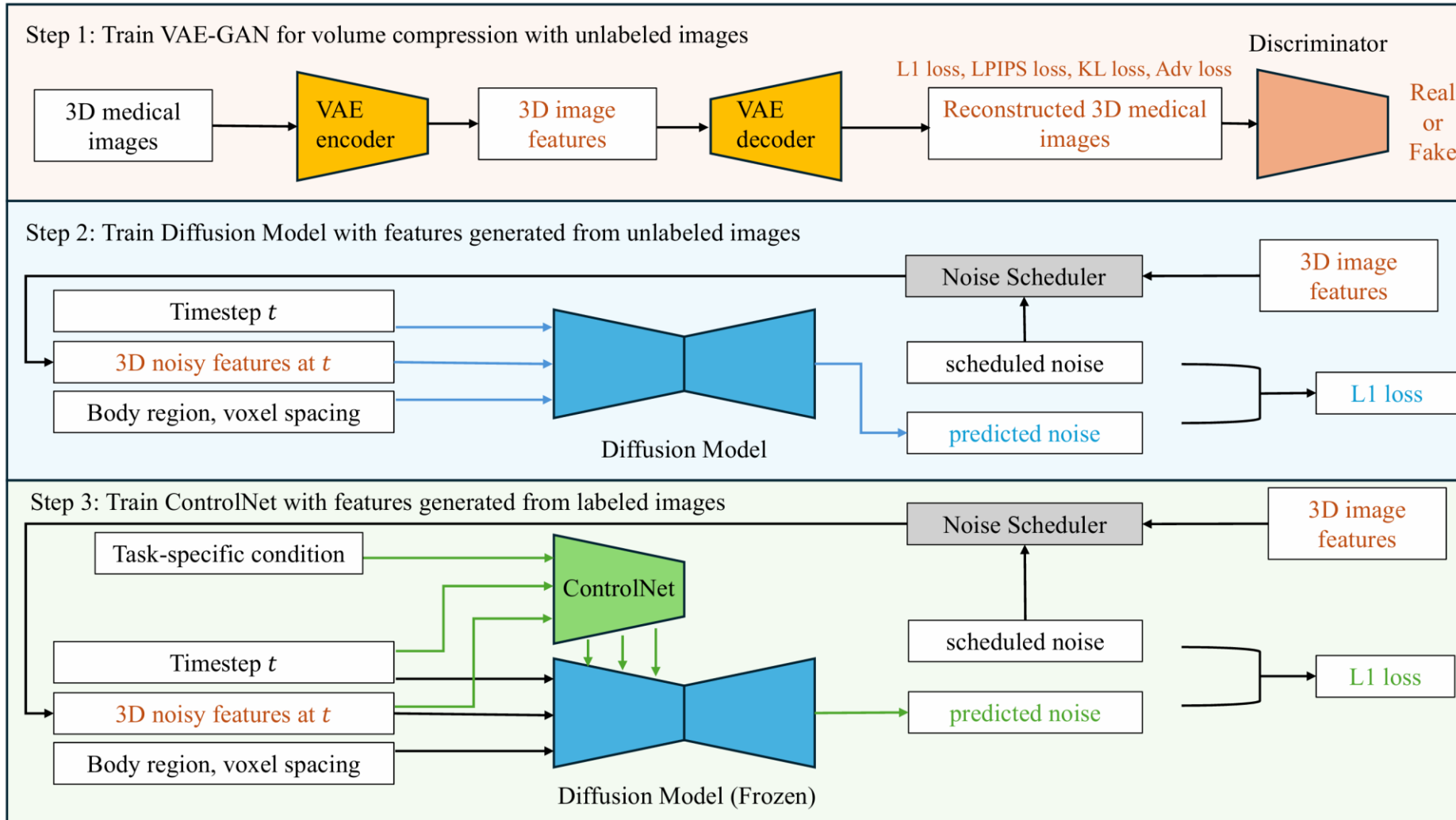
3D Problem : 3D generative model

X-Diffusion: Generating Detailed 3D MRI Volumes From a Single Image Using Cross-Sectional Diffusion Models



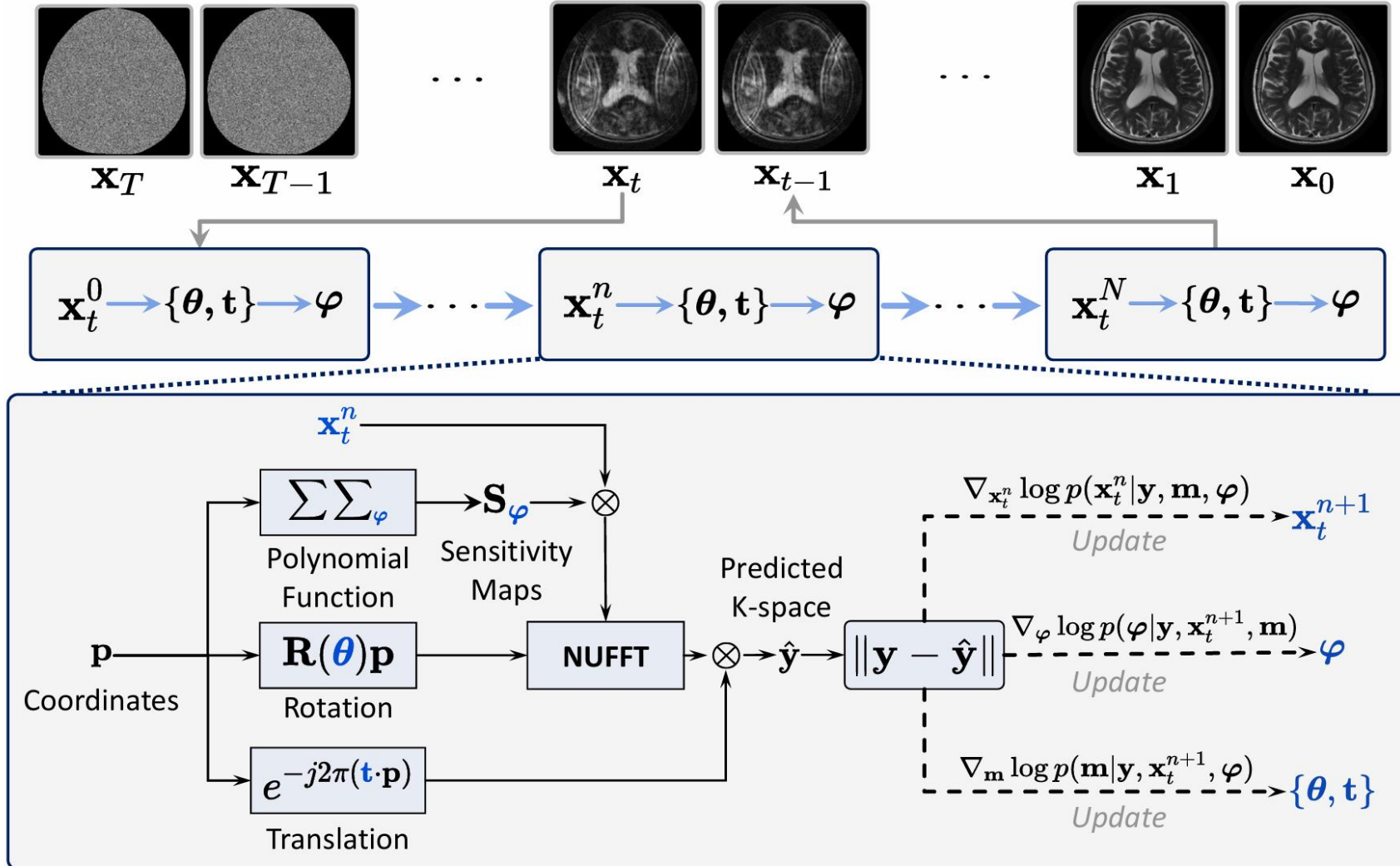
3D Problem : 3D generative model

MAISI, from mask to volume



Fuzzy forward model: *multi-parameter estimation*

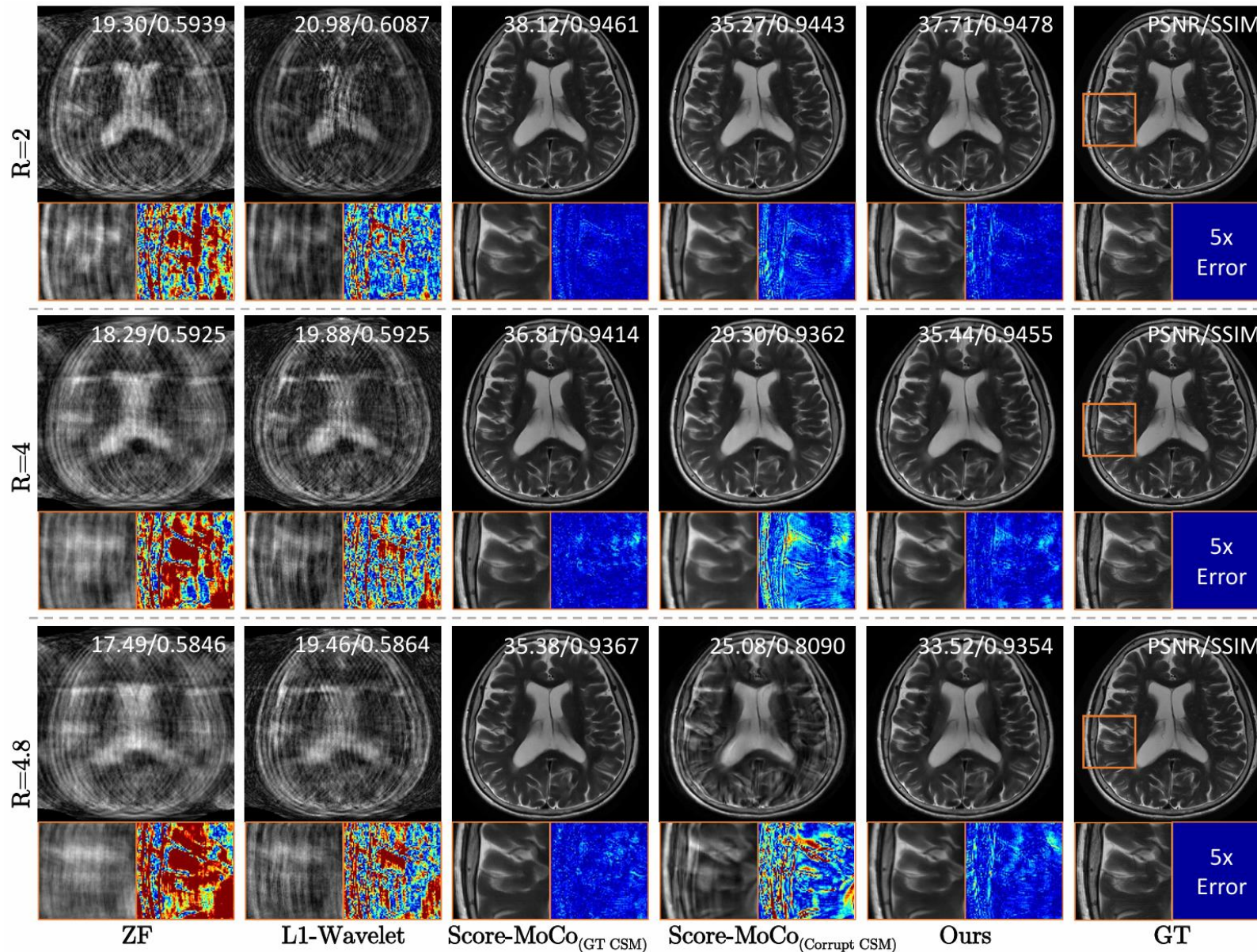
JSMoCo



$$\mathbf{y}_i = \mathbf{PFS}_i \mathbf{x}_0 + \mathbf{n}_i,$$

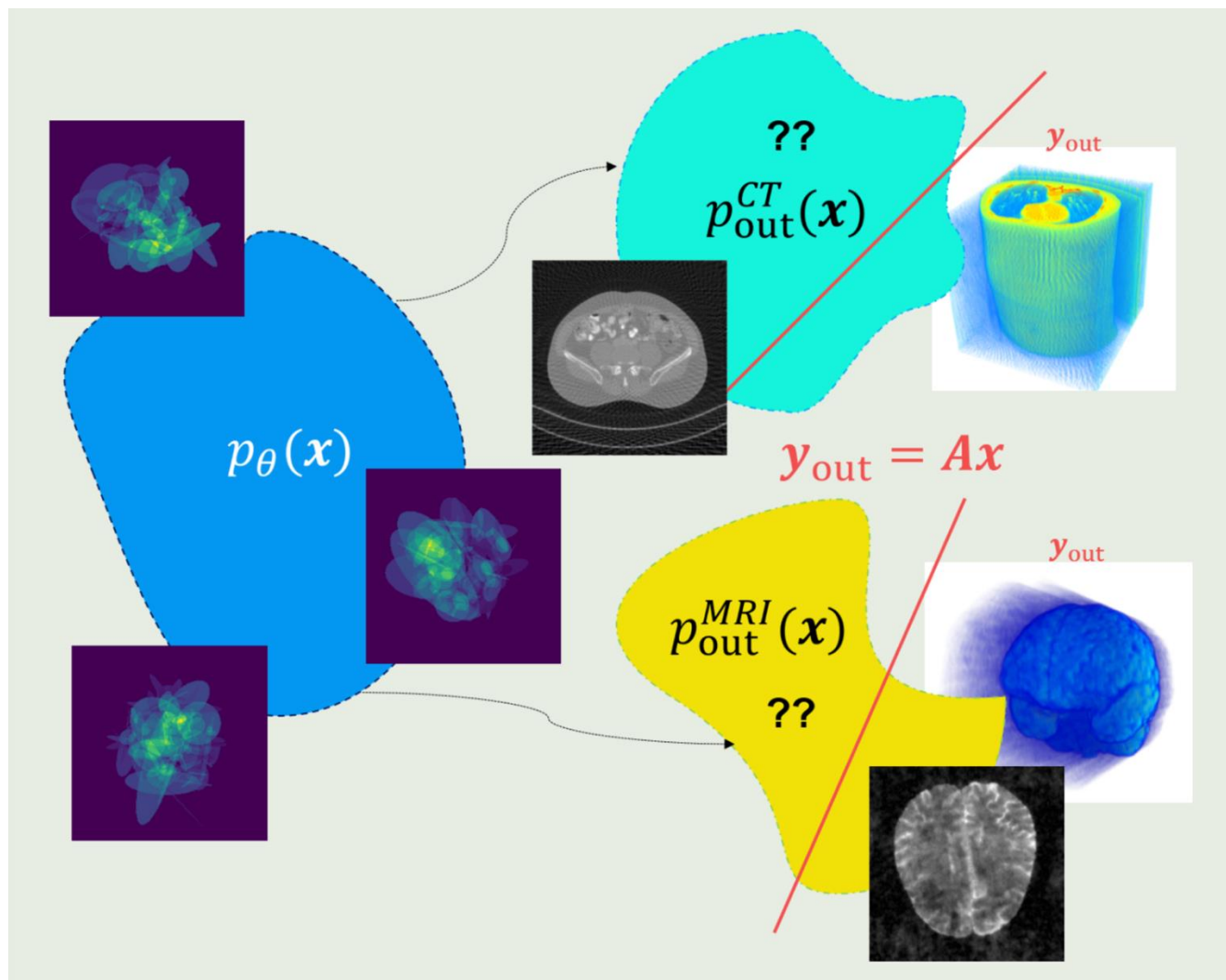
Fuzzy forward model: *multi-parameter estimation*

JSMoCo



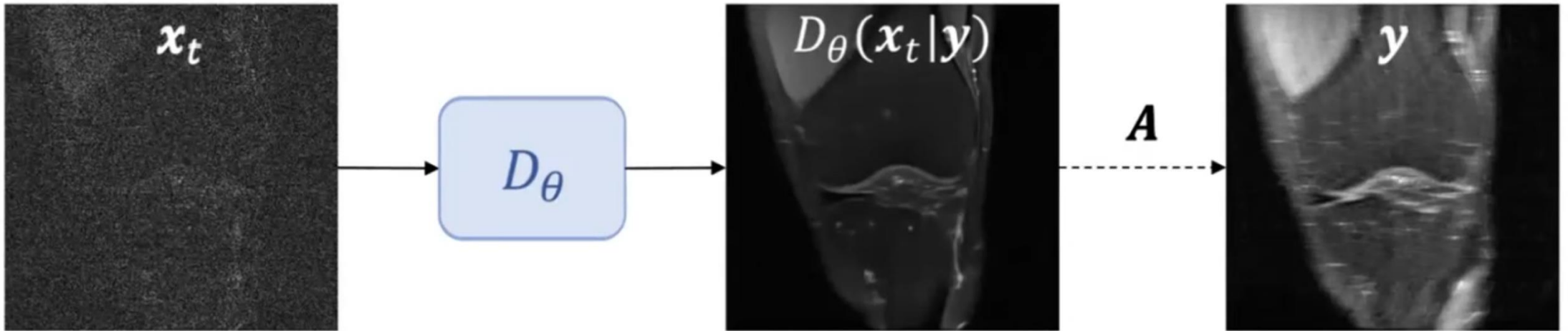
OOD: Combine with DIP or INR

DDIP



OOD: Combine with DIP or INR

DDIP



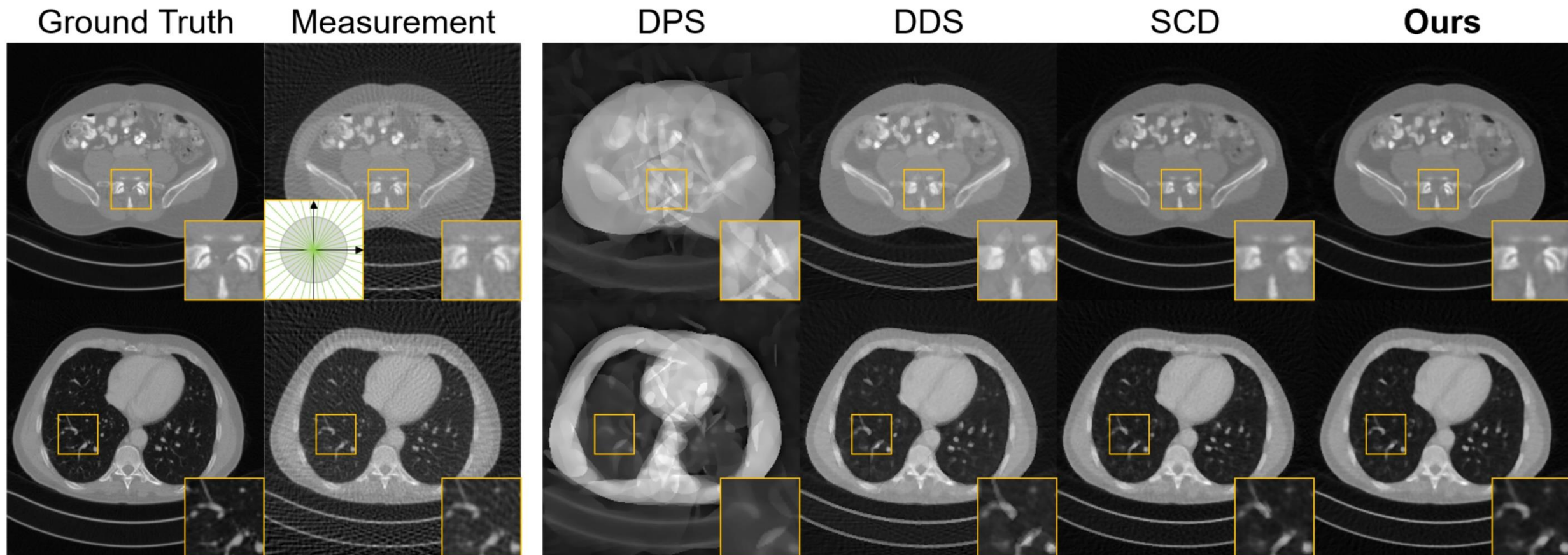
$$\min_{\theta} \|\mathbf{y} - \mathbf{A} D_{\theta}(\mathbf{x}_t | \mathbf{y})\|_2^2$$

$$\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t, \mathbf{y}] \approx D_{\theta}(\mathbf{x}_t | \mathbf{y}) := D_{\theta}(\mathbf{x}_t) + \frac{\sigma_t^2}{2\sigma_y^2} \nabla_{\mathbf{x}_t} \|\mathbf{y} - \mathbf{A} D_{\theta}(\mathbf{x}_t)\|_2^2$$

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DDIP

3D SV-CT



OOD: Combine with DIP or INR

DPER

No data prior, but still a denoiser!

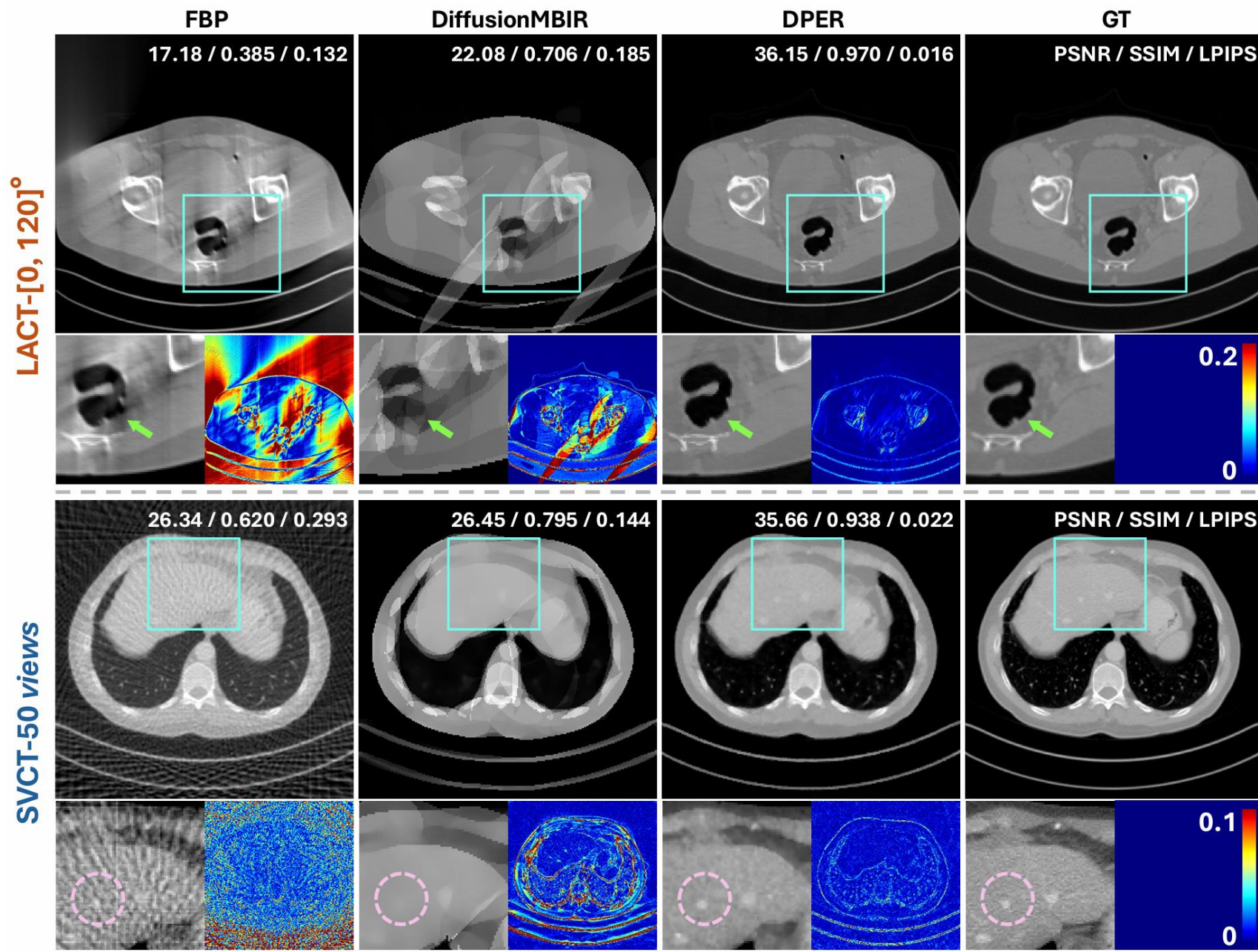
$$\begin{cases} \mathbf{z}_t = \mathbf{x}_{0|t} \approx \mathbf{x}_t + \sigma_t^2 \mathbf{s}_{\theta^*}(\mathbf{x}_t, t), \\ \mathbf{x}_{t-1} = \underset{\mathbf{x}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|^2 + \mu\sigma_i^2 \|\mathbf{x} - \mathbf{x}_{0|t}\|^2 + \sigma_{t-1}\epsilon \end{cases}$$

AWGN denoising

INR-based fidelity refinement

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DPER



Thank you !